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<!DOCTYPE html>
<html lang="en">
<head>
<link rel="preconnect" href="https://fonts.googleapis.com">

<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>

<link
href="https://fonts.googleapis.com/css2?family=Instrument+Serif:ital@0;1&family=Inter:wght
@300;400;500;600&display=swap" rel="stylesheet">
<meta charset="UTF-8"/>
<meta name="viewport" content="width=device-width,initial-scale=1.0"/>
<title>Operation WaveGuard — WAVARA Defense Network</title>
<style>
*,*::before,*::after{box-sizing:border-box;margin:0;padding:0}
:root{
  --bg:#081018;--panel:#111827;--panel2:#1b2a3a;--border:#243447;--border2:#2e4460;
  --accent:#315c67;--blue:#3b82f6;--seafoam:#7dd3c7;--seafoam2:#5bc0be;
  --red:#dc2626;--red2:#ef4444;--amber:#f59e0b;--green:#10b981;
  --text:#e5eef5;--muted:#9fb3c8;--muted2:#7a96ae;--deep:#040c14;
}
html,body{
  margin:0;
  font-family:'Inter',sans-serif;
  color:#e5eef5;
  overflow-x:hidden;

  background:
    radial-gradient(circle at top,
    rgba(59,130,246,.15),
    transparent 35%),

    linear-gradient(
    180deg,
    #081018 0%,
    #111827 45%,
    #0f172a 100%
    );

  min-height:100vh;
}

@keyframes waveScroll{from{background-position-x:0}to{background-position-x:300px}}
@keyframes waveDrift{0%{transform:translateX(0)
scaleY(1)}50%{transform:translateX(-18px) scaleY(1.06)}100%{transform:translateX(0)
scaleY(1)}}
@keyframes
pulseRing{0%{transform:scale(.5);opacity:.9}100%{transform:scale(3);opacity:0}}
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@keyframes radarSweep{from{transform:rotate(0deg)}to{transform:rotate(360deg)}}
@keyframes blinkAlert{0%,100%{opacity:1}50%{opacity:.08}}
@keyframes
shake{0%,100%{transform:translateX(0)}25%{transform:translateX(-8px)}75%{transform:tran
sleteX(8px)}}
@keyframes
fadeInUp{from{opacity:0;transform:translateY(22px)}to{opacity:1;transform:translateY(0)}}
@keyframes fadeIn{from{opacity:0}to{opacity:1}}
@keyframes scanline{0%{top:-4%}100%{top:104%}}
@keyframes starTwinkle{0%,100%{opacity:.12}50%{opacity:.75}}
@keyframes
lockReveal{from{opacity:1;transform:scale(1)}to{opacity:0;transform:scale(1.03)}}
@keyframes
dashIn{from{opacity:0;transform:translateY(7px)}to{opacity:1;transform:translateY(0)}}
@keyframes buoyBob{0%,100%{transform:translateY(0)
rotate(-.8deg)}50%{transform:translateY(-7px) rotate(.8deg)}}
@keyframes insigniaPulse{0%,100%{opacity:.45;filter:drop-shadow(0 0 2px
var(--seafoam2))}50%{opacity:.85;filter:drop-shadow(0 0 10px rgba(91,192,190,.5))}}
@keyframes bootText{from{opacity:0}to{opacity:1}}
@keyframes
redFlash{0%,100%{background:rgba(220,38,38,.04)}50%{background:rgba(220,38,38,.12)}}
@keyframes terminalBlink{0%,49%{opacity:1}50%,100%{opacity:0}}
@keyframes lockdownGlow{0%,100%{box-shadow:inset 0 0 40px
rgba(220,38,38,.06)}50%{box-shadow:inset 0 0 80px rgba(220,38,38,.14)}}

.panel{background:var(--panel);border:1px solid
var(--border);border-radius:9px;padding:16px;position:relative;overflow:hidden}
.panel::before{
content:"";

position:absolute;

top:0;
left:0;
right:0;

height:1px;

background:
linear-gradient(
90deg,
transparent,
rgba(255,255,255,.08),
transparent
);
}
.glass{background:rgba(8,18,30,.82);backdrop-filter:blur(20px);-webkit-backdrop-filter:blur(20
px);border:1px solid rgba(36,52,71,.85);border-radius:12px}

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.glass-light{background:rgba(27,42,58,.5);backdrop-filter:blur(12px);-webkit-backdrop-filter:bl
ur(12px);border:1px solid var(--border);border-radius:9px}
.sec-title{font-size:8.5px;font-weight:700;letter-spacing:3px;text-transform:uppercase;color:va
r(--seafoam);margin-bottom:14px;display:flex;align-items:center;gap:8px}
.sec-title::after{content:"";flex:1;height:1px;background:linear-gradient(90deg,rgba(125,211,19
9,.18),transparent)}
.btn-p{background:transparent;border:1px solid
rgba(91,192,190,.5);color:var(--seafoam);padding:11px
26px;border-radius:6px;font-weight:600;letter-spacing:1.5px;text-transform:uppercase;font-si
ze:11px;transition:all .3s}
.btn-p:hover{background:rgba(91,192,190,.09);box-shadow:0 2px 24px
rgba(91,192,190,.14);transform:translateY(-1px)}
.btn-p:disabled{opacity:.3;cursor:not-allowed;transform:none}
.btn-d{background:transparent;border:1px solid
rgba(239,68,68,.4);color:var(--red2);padding:10px
22px;border-radius:6px;font-weight:600;letter-spacing:1.5px;font-size:11px;transition:all .25s}
.btn-d:hover{background:rgba(220,38,38,.1);box-shadow:0 0 18px rgba(220,38,38,.18)}
.btn-d:disabled{opacity:.3;cursor:not-allowed}
.tab-btn{background:transparent;border:1px solid
var(--border);color:var(--muted2);padding:6px
13px;border-radius:5px;font-size:9px;letter-spacing:1px;text-transform:uppercase;font-weight
:600;transition:all .2s}
.tab-btn:hover{border-color:var(--seafoam2);color:var(--seafoam)}
.tab-btn.active{background:rgba(91,192,190,.1);border-color:var(--seafoam2);color:var(--seaf
oam)}
.badge{display:inline-flex;align-items:center;gap:4px;padding:3px
9px;border-radius:3px;font-size:7.5px;font-weight:700;letter-spacing:2px;text-transform:uppe
rcase}
.lvl-1{background:rgba(16,185,129,.1);border:1px solid
rgba(16,185,129,.28);color:var(--green)}
.lvl-3{background:rgba(59,130,246,.1);border:1px solid
rgba(59,130,246,.28);color:var(--blue)}
.lvl-5{background:rgba(245,158,11,.1);border:1px solid
rgba(245,158,11,.28);color:var(--amber)}
.lvl-sci{background:rgba(125,211,199,.08);border:1px solid
rgba(125,211,199,.28);color:var(--seafoam)}
.bar{height:5px;border-radius:3px;background:rgba(36,52,71,.8);overflow:hidden}
.bar-fill{height:100%;border-radius:3px;transition:width .6s ease;background .5s}
.prog-bar{height:3px;background:var(--border);border-radius:2px;overflow:hidden}
.prog-fill{height:100%;border-radius:2px;transition:width .12s linear}
.gg{background-image:linear-gradient(rgba(36,52,71,.55) 1px,transparent
1px),linear-gradient(90deg,rgba(36,52,71,.55) 1px,transparent 1px);background-size:34px
34px}
.mono{font-family:'Courier New',Courier,monospace}
.page{display:none}.page.active{display:block;animation:fadeIn .3s ease}
.omk-content{display:none}.omk-content.active{display:block}

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```
#navbar{position:sticky;top:0;z-index:200;background:rgba(8,16,24,.93);backdrop-filter:blur(18px);-webkit-backdrop-filter:blur(18px);border-bottom:1px solid
var(--border);display:flex;align-items:center;gap:2px;padding:5px 20px;overflow-x:auto}
.nav-link{background:none;border:none;color:var(--muted2);padding:7px
11px;border-radius:4px;font-size:9px;font-weight:700;letter-spacing:1.5px;text-transform:upper
case;transition:all .2s;white-space:nowrap}
.nav-link:hover{color:var(--seafoam);background:rgba(91,192,190,.06)}
.nav-link.active{color:var(--seafoam);background:rgba(91,192,190,.09)}
.g2{display:grid;grid-template-columns:1fr 1fr;gap:14px}
.g3{display:grid;grid-template-columns:1fr 1fr 1fr;gap:14px}
.g4{display:grid;grid-template-columns:repeat(4,1fr);gap:12px}
@media(max-width:900px){.g3{grid-template-columns:1fr 1fr}.g4{grid-template-columns:1fr
1fr}#sim-grid{grid-template-columns:1fr!important}}
@media(max-width:600px){.g2,.g3,.g4{grid-template-columns:1fr}}
.wrap{max-width:1240px;margin:0 auto;padding:32px 20px}
.sec-head{display:flex;align-items:center;gap:12px;margin-bottom:24px}
.sec-head
h2{color:var(--seafoam);font-size:17px;font-weight:800;letter-spacing:3px;text-transform:upper
case}
.sec-head::after{content:"";flex:1;height:1px;background:linear-gradient(90deg,rgba(125,211,
199,.22),transparent)}
.sci-card{background:linear-gradient(145deg,var(--panel2),var(--panel));border:1px solid
var(--border);border-radius:11px;padding:20px;position:relative;overflow:hidden;transition:all
.3s}
.sci-card:hover{border-color:var(--seafoam2);box-shadow:0 0 22px
rgba(91,192,190,.08);transform:translateY(-2px)}
.s-status{transition:color .3s}
```

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/* — FIXED SCREENS — */
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#lock-screen{position:fixed;inset:0;z-index:9999;overflow:hidden;background:var(--deep)}
#loading-screen{position:fixed;inset:0;z-index:9998;background:var(--bg);display:none;flex-di
rection:column;align-items:center;justify-content:center}
#lockdown-screen{position:fixed;inset:0;z-index:10000;display:none;flex-direction:column;ali
gn-items:center;justify-content:center;background:#040810;animation:lockdownGlow 3s
ease-in-out infinite}
#denial-screen{position:fixed;inset:0;z-index:10001;display:none;flex-direction:column;align-i
tems:center;justify-content:center;background:#000}
#omk-lock-screen{position:fixed;inset:0;z-index:9999;display:none;align-items:center;justify-c
ontent:center;background:rgba(4,8,16,.97)}
#omk-overlay{position:fixed;inset:0;z-index:8000;background:rgba(4,8,16,.97);display:none;o
verflow-y:auto}
</style>
</head>
<body>
```

```
<!-- PERMANENT DENIAL SCREEN -->
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```
<div id="denial-screen">
```

```
<div style="position:absolute;inset:0;animation:redFlash 2.5s ease-in-out infinite"></div>
```

```

<div style="position:relative;z-index:10;text-align:center;padding:0 20px">
  <div style="width:2px;height:60px;background:var(--red);margin:0 auto 28px"></div>
  <div class="mono"
style="color:rgba(239,68,68,.5);font-size:9px;letter-spacing:4px;margin-bottom:12px">WAVA
RA DEFENSE NETWORK — TERMINAL</div>
  <div
style="color:var(--red2);font-size:26px;font-weight:900;letter-spacing:4px;text-transform:uppe
rcase;margin-bottom:10px">ACCESS PERMANENTLY DENIED</div>
  <div class="mono"
style="color:rgba(239,68,68,.4);font-size:10px;letter-spacing:3px;text-transform:uppercase;m
argin-bottom:30px">SESSION TERMINATED</div>
  <div style="width:180px;height:1px;background:rgba(220,38,38,.25);margin:0 auto
28px"></div>
  <div class="mono"
style="color:rgba(239,68,68,.3);font-size:8px;letter-spacing:2px;line-height:2">
    INCIDENT LOGGED — REF: <span id="denial-ref"></span><br/>
    SECURITY TEAM NOTIFIED<br/>
    IP ADDRESS FLAGGED<br/>
    FURTHER ATTEMPTS WILL BE PROSECUTED
  </div>
</div>
</div>
</div>

<!-- LOCKDOWN SCREEN -->
<div id="lockdown-screen">
  <div style="position:absolute;inset:0;background:radial-gradient(ellipse at 50%
40%,#1a0404 0%,#040810 70%);z-index:0"></div>
  <div style="position:absolute;inset:0;background-image:linear-gradient(rgba(220,38,38,.07)
1px,transparent 1px),linear-gradient(90deg,rgba(220,38,38,.07) 1px,transparent
1px);background-size:50px 50px;z-index:1"></div>
  <div id="ld-scanline"
style="position:absolute;left:0;right:0;height:1px;background:linear-gradient(90deg,transpare
nt,rgba(220,38,38,.45),transparent);z-index:2;animation:scanline 2.5s linear infinite"></div>
  <div class="glass" id="ld-panel"
style="position:relative;z-index:10;max-width:480px;width:92%;border-color:rgba(220,38,38,.
35);padding:0;overflow:hidden;animation:fadeInUp .7s ease">
    <div
style="background:linear-gradient(90deg,rgba(220,38,38,.25),rgba(220,38,38,.08));border-b
ottom:1px solid rgba(220,38,38,.3);padding:9px
24px;display:flex;justify-content:space-between;align-items:center">
      <span class="mono"
style="font-size:8px;font-weight:700;letter-spacing:3px;color:rgba(239,68,68,.9);text-transfor
m:uppercase;animation:blinkAlert 1.2s step-end infinite">█ SECURITY LOCKDOWN
ACTIVE</span>
      <span class="mono"
style="font-size:8px;color:rgba(239,68,68,.45);letter-spacing:2px">LD-7</span>
    </div>
  <div style="padding:36px 40px">

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<div style="text-align:center;margin-bottom:28px">
  <div style="width:48px;height:48px;border:1.5px solid
  rgba(220,38,38,.5);border-radius:50%;margin:0 auto
  16px;display:flex;align-items:center;justify-content:center;animation:blinkAlert 1.5s step-end
  infinite">
    <div style="width:20px;height:20px;border:1.5px solid
    var(--red);border-radius:50%;display:flex;align-items:center;justify-content:center"><div
    style="width:6px;height:6px;background:var(--red2);border-radius:50%"></div></div>
    </div>
    <div
    style="font-size:8px;letter-spacing:4px;color:rgba(239,68,68,.55);text-transform:uppercase;m
    argin-bottom:8px">Authentication Failure — Maximum Attempts Exceeded</div>
    <div
    style="font-size:20px;font-weight:800;color:var(--red2);letter-spacing:2px;text-transform:uppe
    rcase;margin-bottom:5px">Level 5 Clearance Required</div>
    <div
    style="font-size:9px;color:rgba(239,68,68,.4);letter-spacing:1.5px;text-transform:uppercase">
    Override access code required. One attempt permitted.</div>
    </div>
    <label
    style="display:block;font-size:8px;letter-spacing:2.5px;color:rgba(239,68,68,.5);text-transfor
    m:uppercase;margin-bottom:8px">Level 5 Override Code</label>
    <input type="password" id="ld-input" placeholder="● ● ● ● ●" autocomplete="off"
    style="text-align:center;font-size:17px;border-color:rgba(220,38,38,.4)"
    onkeydown="if(event.key==='Enter')checkLv5()"/>
    <div id="ld-error" style="min-height:18px;margin:8px
    0;text-align:center;font-size:9px;color:var(--red2);letter-spacing:1px;opacity:0;transition:opaci
    ty .2s"></div>
    <button onclick="checkLv5()"
    style="width:100%;padding:13px;background:linear-gradient(135deg,rgba(220,38,38,.2),rgb
    a(220,38,38,.06));border:1px solid
    rgba(239,68,68,.4);color:var(--red2);border-radius:6px;font-size:11px;font-weight:700;letter-s
    pacing:2.5px;text-transform:uppercase;cursor:pointer;margin-top:4px">Verify Level 5
    Override</button>
    </div>
  </div>
</div>

<!-- LOCK SCREEN -->
<div id="lock-screen">
  <div id="ls-stars" style="position:absolute;inset:0;pointer-events:none;z-index:1"></div>
  <div style="position:absolute;inset:0;background:radial-gradient(ellipse at 50%
  38%,#0c1e36 0%,#040c16 58%,#020810 100%);z-index:0"></div>
  <div style="position:absolute;inset:0;background-image:linear-gradient(rgba(36,52,71,.14)
  1px,transparent 1px),linear-gradient(90deg,rgba(36,52,71,.14) 1px,transparent
  1px);background-size:58px 58px;z-index:0;opacity:.8"></div>
  <!-- Radar (background, subtle) -->

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<div
style="position:absolute;top:50%;left:50%;transform:translate(-50%,-54%);pointer-events:none;z-index:1;opacity:.07">
  <svg width="640" height="640" viewBox="0 0 640 640"><circle cx="320" cy="320" r="300"
fill="none" stroke="var(--seafoam)" stroke-width=".8"/><circle cx="320" cy="320" r="200"
fill="none" stroke="var(--seafoam)" stroke-width=".6"/><circle cx="320" cy="320" r="100"
fill="none" stroke="var(--seafoam)" stroke-width=".5"/><line x1="20" y1="320" x2="620"
y2="320" stroke="rgba(125,211,199,.4)" stroke-width=".4"/><line x1="320" y1="20" x2="320"
y2="620" stroke="rgba(125,211,199,.4)" stroke-width=".4"/><line x1="320" y1="320"
x2="320" y2="22" stroke="rgba(125,211,199,.9)" stroke-width="1.2"
style="transform-origin:320px 320px;animation:radarSweep 8s linear infinite"/></svg>
</div>
<!-- Ocean waves -->
<svg style="position:absolute;bottom:0;left:0;width:100%;z-index:2" viewBox="0 0 1440
300" preserveAspectRatio="none" height="270">
  <defs>
    <linearGradient id="wg1" x1="0" y1="0" x2="0" y2="1"><stop offset="0%"
stop-color="#0e2e4c" stop-opacity=".65"/><stop offset="100%" stop-color="#030b16"
stop-opacity="1"/></linearGradient>
    <linearGradient id="wg2" x1="0" y1="0" x2="0" y2="1"><stop offset="0%"
stop-color="#183d60" stop-opacity=".38"/><stop offset="100%" stop-color="#020c18"
stop-opacity="1"/></linearGradient>
  </defs>
  <path fill="url(#wg1)" d="M0,108 C240,62 480,172 720,108 C960,44 1200,152 1440,108
L1440,300 L0,300 Z" style="animation:waveDrift 7s ease-in-out infinite"/>
  <path fill="url(#wg2)" d="M0,138 C320,86 640,198 960,138 C1120,96 1300,158 1440,130
L1440,300 L0,300 Z" style="animation:waveDrift 9s ease-in-out infinite reverse"/>
  <path fill="rgba(125,211,199,.04)" d="M0,165 C480,118 960,212 1440,165 L1440,300
L0,300 Z"/>
</svg>
<div id="ls-scanline"
style="position:absolute;left:0;right:0;height:1px;background:linear-gradient(90deg,transparent,rgba(125,211,199,.32),transparent);z-index:3;pointer-events:none;animation:scanline 5s
linear infinite"></div>

<div id="lock-panel" class="glass"
style="position:relative;z-index:10;max-width:480px;width:92%;animation:fadeInUp .9s
ease;padding:0;overflow:hidden">
  <!-- Top bar -->
  <div
style="background:linear-gradient(90deg,rgba(220,38,38,.15),rgba(220,38,38,.04));border-bottom:1px solid rgba(220,38,38,.2);padding:8px
24px;display:flex;justify-content:space-between;align-items:center">
    <span class="mono"
style="font-size:7.5px;font-weight:700;letter-spacing:3px;color:rgba(239,68,68,.75);text-transform:uppercase">CLASSIFIED / RESTRICTED</span>

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    <span class="mono"
style="font-size:7.5px;color:rgba(239,68,68,.4);letter-spacing:2px">ALPHA-7
NETWORK</span>
</div>
<div style="padding:36px 44px 38px">
    <!-- Oceanic insignia -->
    <div style="display:flex;flex-direction:column;align-items:center;margin-bottom:28px">
        <svg width="52" height="52" viewBox="0 0 52 52" fill="none"
style="color:var(--seafoam2);animation:insigniaPulse 4s ease-in-out
infinite;margin-bottom:14px">
            <circle cx="26" cy="26" r="24" stroke="currentColor" stroke-width="1.2" opacity=".7"/>
            <circle cx="26" cy="26" r="15" stroke="currentColor" stroke-width=".7" opacity=".35"/>
            <path d="M4,26 C10,20 18,32 26,26 C34,20 42,32 48,26" stroke="currentColor"
stroke-width="1.8" stroke-linecap="round"/>
        </svg>
        <div class="mono"
style="font-size:7.5px;letter-spacing:5px;color:var(--muted2);font-weight:600;text-transform:u
ppercase;margin-bottom:6px">WAVARA DEFENSE NETWORK</div>
    </div>
    <div style="font-size:21px;font-weight:900;letter-spacing:3px;text-transform:uppercase;backgroun
d:linear-gradient(135deg,var(--seafoam),#d8f5f2
50%,var(--seafoam2));-webkit-background-clip:text;-webkit-text-fill-color:transparent;margin-
bottom:3px">WAVEGUARD</div>
    <div class="mono"
style="font-size:8px;letter-spacing:3.5px;color:var(--muted2);text-transform:uppercase;margi
n-bottom:4px">Hybrid Early Warning System</div>
    <div class="mono"
style="font-size:7.5px;letter-spacing:2px;color:rgba(239,68,68,.55);text-transform:uppercase;
font-weight:700">Authorised Personnel Only</div>
</div>
<!-- Status row -->
<div style="display:flex;justify-content:center;gap:14px;margin-bottom:22px">
    <div style="display:flex;align-items:center;gap:4px"><div
style="width:5px;height:5px;border-radius:50%;background:var(--green);box-shadow:0 0 5px
var(--green)"></div><span class="mono"
style="font-size:7.5px;color:var(--muted2);letter-spacing:1.5px;text-transform:uppercase">Sy
stems</span></div>
    <div style="display:flex;align-items:center;gap:4px"><div
style="width:5px;height:5px;border-radius:50%;background:var(--amber);box-shadow:0 0
5px var(--amber);animation:blinkAlert 1.5s step-end infinite"></div><span class="mono"
style="font-size:7.5px;color:var(--muted2);letter-spacing:1.5px;text-transform:uppercase">Au
th Pending</span></div>
    <div style="display:flex;align-items:center;gap:4px"><div
style="width:5px;height:5px;border-radius:50%;background:var(--blue);box-shadow:0 0 5px
var(--blue)"></div><span class="mono"
style="font-size:7.5px;color:var(--muted2);letter-spacing:1.5px;text-transform:uppercase">Se
nsors</span></div>
</div>

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    <!-- Attempt indicator -->
    <div id="attempt-row"
style="display:flex;justify-content:center;gap:6px;margin-bottom:16px"></div>
    <label class="mono"
style="display:block;font-size:7.5px;letter-spacing:2.5px;color:var(--muted2);text-transform:u
ppercase;margin-bottom:8px">Security Access Code</label>
    <input type="password" id="lock-input" placeholder="● ● ● ● ●" autocomplete="off"
style="text-align:center;font-size:18px;letter-spacing:8px"
onkeydown="if(event.key==='Enter')checkLock()"/>
    <div id="lock-error" style="min-height:18px;margin:8px
0;text-align:center;font-size:9px;color:var(--red2);letter-spacing:1px;opacity:0;transition:opaci
ty .2s;font-weight:600"></div>
    <button onclick="checkLock()"
style="width:100%;padding:14px;background:linear-gradient(135deg,rgba(49,92,103,.38),rg
ba(59,130,246,.15));border:1px solid
rgba(91,192,190,.45);color:var(--seafoam);border-radius:6px;font-size:11px;font-weight:700;l
etter-spacing:3px;text-transform:uppercase;cursor:pointer;transition:all .3s"
onmouseover="this.style.background='linear-gradient(135deg,rgba(91,192,190,.25),rgba(59,
130,246,.22))'"
onmouseout="this.style.background='linear-gradient(135deg,rgba(49,92,103,.38),rgba(59,13
0,246,.15))'">Authenticate Access</button>
    <div class="mono"
style="margin-top:16px;text-align:center;font-size:8px;color:rgba(36,52,71,.9);letter-spacing:
1.5px">ENC-AES256 · SYS v2.4.1 · WAVARA-NET</div>
  </div>
</div>
</div>

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<!-- LOADING SCREEN -->
<div id="loading-screen">
  <div style="position:absolute;inset:0;background-image:linear-gradient(rgba(36,52,71,.12)
1px,transparent 1px),linear-gradient(90deg,rgba(36,52,71,.12) 1px,transparent
1px);background-size:42px 42px"></div>
  <div
style="position:absolute;left:0;right:0;height:1px;background:linear-gradient(90deg,transpare
nt,rgba(125,211,199,.3),transparent);animation:scanline 3s linear infinite"></div>
  <div style="position:relative;z-index:10;width:100%;max-width:520px;padding:0
24px;text-align:center">
    <!-- Radar animation -->
    <div style="display:flex;justify-content:center;margin-bottom:24px">
      <svg width="82" height="82" viewBox="0 0 82 82">
        <circle cx="41" cy="41" r="38" fill="none" stroke="rgba(125,211,199,.1)"
stroke-width="1"/>
        <circle cx="41" cy="41" r="25" fill="none" stroke="rgba(125,211,199,.14)"
stroke-width="1"/>
        <circle cx="41" cy="41" r="12" fill="none" stroke="rgba(125,211,199,.18)"
stroke-width="1"/>
        <circle cx="41" cy="41" r="3" fill="var(--seafoam2)"/>
      </svg>
    </div>
  </div>

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    <line x1="41" y1="41" x2="41" y2="4" stroke="rgba(125,211,199,.55)"
stroke-width="1.5" style="transform-origin:41px 41px;animation:radarSweep 2.2s linear
infinite"/>
    <path d="M41 41 L41 4 A37 37 0 0 1 68 68 Z" fill="rgba(125,211,199,.04)"/>
    <circle cx="41" cy="41" r="0" fill="none" stroke="rgba(125,211,199,.35)"
stroke-width="1.8" style="animation:pulseRing 2.2s ease-out infinite"/>
    <circle cx="41" cy="41" r="0" fill="none" stroke="rgba(125,211,199,.2)"
stroke-width="1.2" style="animation:pulseRing 2.2s ease-out infinite;animation-delay:.7s"/>
  </svg>
</div>
<div class="mono"
style="font-size:8px;letter-spacing:4px;color:var(--muted2);text-transform:uppercase;margin-
bottom:5px">WAVARA DEFENSE NETWORK</div>
<div
style="font-size:26px;font-weight:900;letter-spacing:5px;color:var(--seafoam);text-transform:
uppercase;margin-bottom:4px">WAVEGUARD</div>
<div class="mono"
style="font-size:8px;letter-spacing:3px;color:var(--muted2);text-transform:uppercase;margin-
bottom:28px">Hybrid Tsunami Early Warning System</div>
<!-- Boot log -->
<div id="boot-log" class="mono"
style="text-align:left;background:rgba(4,12,20,.7);border:1px solid
var(--border);border-radius:6px;padding:14px;margin-bottom:18px;height:90px;overflow:hidd
en;font-size:9px;color:var(--seafoam2);line-height:2"></div>
<div class="prog-bar"><div id="load-bar" class="prog-fill"
style="width:0;background:linear-gradient(90deg,var(--accent),var(--seafoam))"></div></div>
<div style="display:flex;justify-content:space-between;margin-top:7px">
  <span id="load-phase" class="mono"
style="font-size:9px;color:var(--muted2)">Standby...</span>
  <span id="load-pct" class="mono" style="font-size:9px;color:var(--seafoam)">0%</span>
</div>
</div>
</div>

<!-- OMK LOCK -->
<div id="omk-lock-screen">
  <div style="position:absolute;inset:0;background:radial-gradient(ellipse at 50%
40%,#180404 0%,#040810 80%);z-index:0"></div>
  <div style="position:absolute;inset:0;background-image:linear-gradient(rgba(220,38,38,.05)
1px,transparent 1px),linear-gradient(90deg,rgba(220,38,38,.05) 1px,transparent
1px);background-size:50px 50px;z-index:1"></div>
  <div class="glass" id="omk-lock-panel"
style="position:relative;z-index:10;max-width:420px;width:92%;border-color:rgba(220,38,38,.
28);padding:0;overflow:hidden;animation:fadeInUp .6s ease">
    <div
style="background:linear-gradient(90deg,rgba(220,38,38,.2),rgba(220,38,38,.05));border-bot
tom:1px solid rgba(220,38,38,.28);padding:8px
24px;display:flex;justify-content:space-between">

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    <span class="mono"
style="font-size:7.5px;font-weight:700;letter-spacing:3px;color:rgba(239,68,68,.85);text-transf
orm:uppercase">△ RESTRICTED ACCESS</span>
    <span class="mono"
style="font-size:7.5px;color:rgba(239,68,68,.45);letter-spacing:2px">LEVEL 5</span>
</div>
<div style="padding:34px 40px 36px">
    <div style="text-align:center;margin-bottom:22px">
        <div style="font-size:30px;margin-bottom:10px">🔒</div>
        <div class="mono"
style="font-size:7.5px;letter-spacing:4px;color:rgba(239,68,68,.55);font-weight:700;text-transf
orm:uppercase;margin-bottom:5px">OMK Intelligence System</div>
        <div
style="font-size:17px;font-weight:800;letter-spacing:2px;color:var(--red2);text-transform:uppe
rcase;margin-bottom:3px">Classified Access</div>
        <div class="mono"
style="font-size:8px;color:rgba(239,68,68,.38);letter-spacing:1.5px;text-transform:uppercase"
>Unauthorised access prosecuted under WAV-7</div>
    </div>
    <label class="mono"
style="display:block;font-size:7.5px;letter-spacing:2.5px;color:rgba(239,68,68,.5);text-transfo
rm:uppercase;margin-bottom:7px">Classified Authorisation Code</label>
    <input type="password" id="omk-input" placeholder="● ● ● ● ● ●" autocomplete="off"
style="text-align:center;font-size:17px;letter-spacing:8px;border-color:rgba(220,38,38,.38)"
onkeydown="if(event.key==='Enter')checkOmk()"/>
    <div id="omk-error" style="min-height:17px;margin:8px
0;text-align:center;font-size:9px;color:var(--red2);letter-spacing:1px;opacity:0;transition:opaci
ty .2s"></div>
    <button onclick="checkOmk()"
style="width:100%;padding:13px;background:linear-gradient(135deg,rgba(220,38,38,.18),rg
ba(220,38,38,.06));border:1px solid
rgba(239,68,68,.4);color:var(--red2);border-radius:6px;font-size:11px;font-weight:700;letter-s
pacing:2.5px;text-transform:uppercase;cursor:pointer;margin-bottom:9px">Access Classified
System</button>
    <button onclick="closeOmkLock()"
style="width:100%;padding:9px;background:transparent;border:1px solid
var(--border);color:var(--muted2);border-radius:5px;font-size:9px;cursor:pointer">Cancel</bu
tton>
    </div>
</div>
</div>

<!-- OMK OVERLAY -->
<div id="omk-overlay">
    <div style="max-width:1200px;margin:0 auto;padding:32px 20px">
        <div
style="display:flex;align-items:center;justify-content:space-between;margin-bottom:24px;pad

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ding:14px 20px;background:rgba(220,38,38,.05);border:1px solid
rgba(220,38,38,.18);border-radius:9px">
  <div>
    <div class="mono"
style="font-size:7.5px;letter-spacing:4px;color:rgba(239,68,68,.6);font-weight:700;text-transfo
rm:uppercase;margin-bottom:4px">🔒 CLASSIFIED — OMK INTELLIGENCE —
OPERATION WAVEGUARD</div>
    <div
style="font-size:18px;font-weight:800;letter-spacing:2px;color:var(--red2);text-transform:uppe
rcase">OMK Command Dashboard</div>
    </div>
    <button onclick="closeOmk()" class="tab-btn"
style="border-color:rgba(239,68,68,.35);color:var(--red2);flex-shrink:0">X Secure
Close</button>
  </div>
  <div style="display:flex;gap:7px;flex-wrap:wrap;margin-bottom:20px">
    <button class="tab-btn active" id="omk-t-0" onclick="setOmkTab(0)"
style="border-color:rgba(239,68,68,.35);color:var(--red2)">Complete Guide</button>
    <button class="tab-btn" id="omk-t-1" onclick="setOmkTab(1)">Science Notes</button>
    <button class="tab-btn" id="omk-t-2" onclick="setOmkTab(2)">Presentation
Script</button>
    <button class="tab-btn" id="omk-t-3" onclick="setOmkTab(3)">Teacher
Q&A</button>
    <button class="tab-btn" id="omk-t-4" onclick="setOmkTab(4)">Quick Summary</button>
  </div>
  <!-- GUIDE -->
  <div class="omk-content active" id="omk-c-0">
    <div class="g2">
      <div class="panel">
        <div class="sec-title" style="color:var(--red2)">Section-by-Section Explanation</div>
        <div
style="display:flex;flex-direction:column;gap:10px;font-size:11.5px;color:var(--text);line-height
:1.75">
          <div
style="padding:10px;background:rgba(125,211,199,.04);border-radius:6px;border-left:2px
solid var(--seafoam2)"><strong style="color:var(--seafoam2)">Lock Screen:</strong>
Simulates a real government access system. In NOAA/DART networks, stations are
restricted. The 3-attempt lockdown mirrors real security protocols.</div>
          <div
style="padding:10px;background:rgba(125,211,199,.04);border-radius:6px;border-left:2px
solid var(--seafoam2)"><strong style="color:var(--seafoam2)">Key Stats (7000km / 14000
km/h / 1000 km/h / 6.5h):</strong> Realistic Pacific Ocean values. Japan-California = ~8900
km. P-waves travel 5–14 km/s through crust. This gives a 6–7 hour warning
advantage.</div>
          <div
style="padding:10px;background:rgba(125,211,199,.04);border-radius:6px;border-left:2px
solid var(--seafoam2)"><strong style="color:var(--seafoam2)">Simulation:</strong> 5 views of

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the same event. The timeline bar is the most important visual: seismic detection at 30 min vs tsunami arrival at 7 hrs = 6.5 hour gap.</div>

<div style="padding:10px;background:rgba(125,211,199,.04);border-radius:6px;border-left:2px solid var(--seafoam2)"><strong style="color:var(--seafoam)">Radar Comparison: The green hybrid polygon always covers the largest area across all 5 axes. Visual proof of superiority.</div>

<div style="padding:10px;background:rgba(125,211,199,.04);border-radius:6px;border-left:2px solid var(--seafoam2)"><strong style="color:var(--seafoam)">Wave Science: Hover the concept cards. Each covers GCSE Physics waves material: amplitude, wavelength, frequency, transverse vs longitudinal.</div>

</div>
</div>
<div class="panel">
 <div class="sec-title" style="color:var(--red2)">Key Numbers</div>
 <div style="display:flex;flex-direction:column;gap:10px">
 <div style="padding:12px;background:rgba(59,130,246,.06);border:1px solid rgba(59,130,246,.18);border-radius:7px"><div class="mono" style="font-size:7.5px;color:var(--muted2);letter-spacing:2px;text-transform:uppercase;margin-bottom:3px">Distance</div><div class="mono" style="color:var(--blue);font-size:22px;font-weight:700">7,000 km</div><div style="font-size:9.5px;color:var(--muted2);margin-top:2px">Realistic Pacific Ocean epicentre-to-coast scale</div></div>
 <div style="padding:12px;background:rgba(245,158,11,.06);border:1px solid rgba(245,158,11,.18);border-radius:7px"><div class="mono" style="font-size:7.5px;color:var(--muted2);letter-spacing:2px;text-transform:uppercase;margin-bottom:3px">Seismic P-Wave Speed</div><div class="mono" style="color:var(--amber);font-size:22px;font-weight:700">14,000 km/h</div><div style="font-size:9.5px;color:var(--muted2);margin-top:2px">Arrives at sensors in 30 minutes</div></div>
 <div style="padding:12px;background:rgba(125,211,199,.05);border:1px solid rgba(125,211,199,.18);border-radius:7px"><div class="mono" style="font-size:7.5px;color:var(--muted2);letter-spacing:2px;text-transform:uppercase;margin-bottom:3px">Tsunami Speed</div><div class="mono" style="color:var(--seafoam);font-size:22px;font-weight:700">1,000 km/h</div><div style="font-size:9.5px;color:var(--muted2);margin-top:2px"> $v = \sqrt{(9.81 \times 10,200\text{m})} \approx 316 \text{ m/s} \approx 1,000 \text{ km/h}$ </div></div>
 <div style="padding:12px;background:rgba(16,185,129,.06);border:1px solid rgba(16,185,129,.18);border-radius:7px"><div class="mono" style="font-size:7.5px;color:var(--muted2);letter-spacing:2px;text-transform:uppercase;margin-bottom:3px">Hybrid Advantage</div><div class="mono" style="color:var(--green);font-size:22px;font-weight:700">6.5 hours</div><div style="font-size:9.5px;color:var(--muted2);margin-top:2px">7.0h arrival – 0.5h detection = 6.5h evacuation window</div></div>
 </div>
</div>
</div>

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</div>
<!-- SCIENCE -->
<div class="omk-content" id="omk-c-1">
  <div class="g2">
    <div class="panel">
      <div class="sec-title" style="color:var(--red2)">Wave Physics</div>
      <div
style="display:flex;flex-direction:column;gap:11px;font-size:11.5px;line-height:1.75">
        <div
style="padding:10px;background:rgba(125,211,199,.04);border-radius:6px;border-left:2px
solid var(--seafoam)"><div
style="color:var(--seafoam);font-weight:700;margin-bottom:3px;font-size:10px">Amplitude</d
iv>"Deep-ocean tsunami amplitude is just 0.3–1m — invisible to ships. Near shore, energy
conservation (shoaling) amplifies it to 10–40m."</div>
        <div
style="padding:10px;background:rgba(125,211,199,.04);border-radius:6px;border-left:2px
solid var(--seafoam)"><div
style="color:var(--seafoam);font-weight:700;margin-bottom:3px;font-size:10px">Wavelength
(λ)</div>"Tsunami λ = 200–500 km. Wind wave λ ≈ 100m. Long wavelength = full water
column moves when seabed shifts = catastrophic energy."</div>
        <div
style="padding:10px;background:rgba(125,211,199,.04);border-radius:6px;border-left:2px
solid var(--amber)"><div
style="color:var(--amber);font-weight:700;margin-bottom:3px;font-size:10px">v = fλ & v =
√(gd)</div>"At depth 4000m: v = √(9.81×4000) = 198 m/s = 713 km/h. At 10,000m: v =
√(9.81×10000) = 313 m/s = 1,127 km/h."</div>
        <div
style="padding:10px;background:rgba(125,211,199,.04);border-radius:6px;border-left:2px
solid var(--blue)"><div
style="color:var(--blue);font-weight:700;margin-bottom:3px;font-size:10px">P-waves vs
S-waves</div>"P-waves are longitudinal compression waves — travel fastest through rigid
rock at 14,000 km/h. S-waves are transverse, slower, can't travel through liquids."</div>
        <div
style="padding:10px;background:rgba(125,211,199,.04);border-radius:6px;border-left:2px
solid var(--green)"><div
style="color:var(--green);font-weight:700;margin-bottom:3px;font-size:10px">Energy
Transfer</div>"Waves transfer energy, not matter. Buoys bob in place as a 1000 km/h
tsunami passes — the energy travels, the water particles barely move horizontally."</div>
      </div>
    </div>
    <div class="panel">
      <div class="sec-title" style="color:var(--red2)">Why Hybrid Wins</div>
      <div
style="display:flex;flex-direction:column;gap:11px;font-size:11.5px;line-height:1.75">
        <div style="padding:11px;background:rgba(49,92,103,.1);border-radius:6px"><div
style="color:var(--amber);font-weight:700;margin-bottom:3px;font-size:10px">⚡ Seismic
Only — Problem</div>Not every M6+ earthquake generates a tsunami. Seismographs alone

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would cause constant false alarms. 1946 Hilo false alarm destroyed public trust — people stopped evacuating.</div>

<div style="padding:11px;background:rgba(49,92,103,.1);border-radius:6px"><div style="color:var(--blue);font-weight:700;margin-bottom:3px;font-size:10px"> Buoy Only — Problem</div>DART buoys are highly accurate, but the tsunami wave must physically reach the buoy. At 7000 km, that takes hours — destroying the warning advantage.</div>

<div style="padding:11px;background:rgba(16,185,129,.07);border:1px solid rgba(16,185,129,.14);border-radius:6px"><div style="color:var(--green);font-weight:700;margin-bottom:3px;font-size:10px">✓ Hybrid — Solution</div>Seismic gives speed. Buoy gives confirmation accuracy. Two-factor verification: 85% fewer false alarms + 6.5 hours earlier. Both must agree before full alert.</div>

<div style="padding:10px;background:rgba(59,130,246,.05);border-radius:6px"><div class="mono" style="color:var(--blue);font-size:10px;line-height:1.9">Distance: 7,000 km
Seismic: 14,000 km/h → 30 min
Tsunami: 1,000 km/h → 7 hrs
Advantage: <strong style="color:var(--green)">6.5 hours</div></div>

</div>

</div>

</div>

</div>

<!-- SCRIPT -->

<div class="omk-content" id="omk-c-2">

<div class="g2">

<div class="panel">

<div class="sec-title" style="color:var(--red2)">Presentation Script</div>

<div

style="display:flex;flex-direction:column;gap:10px;font-size:11.5px;line-height:1.75">

<div style="padding:10px;background:rgba(59,130,246,.06);border-left:3px solid var(--blue);border-radius:0 6px 6px 0"><strong style="color:var(--blue)">Opening: "Before accessing our monitoring network, you need security clearance — simulating real government emergency systems." [Enter WAVARA]</div>

<div style="padding:10px;background:rgba(125,211,199,.06);border-left:3px solid var(--seafoam);border-radius:0 6px 6px 0"><strong style="color:var(--seafoam)">Hero Stats: "These four numbers tell the whole story. 7000 km, 14000 km/h seismic, 1000 km/h tsunami. That gives us 6.5 extra hours of warning."</div>

<div style="padding:10px;background:rgba(245,158,11,.06);border-left:3px solid var(--amber);border-radius:0 6px 6px 0"><strong style="color:var(--amber)">Simulation: "The orange pulse is the seismic wave travelling underground at 14,000 km/h. The blue wave is the tsunami on the surface. Watch the timeline bar — that gap is 6.5 hours."</div>

<div style="padding:10px;background:rgba(16,185,129,.06);border-left:3px solid var(--green);border-radius:0 6px 6px 0"><strong style="color:var(--green)">Radar Chart: "The green polygon always covers the most area. On every single axis. That's the data — not an opinion."</div>

<div style="padding:10px;background:rgba(49,92,103,.1);border-left:3px solid var(--accent);border-radius:0 6px 6px 0"><strong style="color:var(--seafoam2)">Closing: "Close on Public Safety. This is what WaveGuard is actually for — not the physics. The people."</div>

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    </div>
  </div>
  <div class="panel">
    <div class="sec-title" style="color:var(--red2)">Interactive Feature Callouts</div>
    <div
style="display:flex;flex-direction:column;gap:10px;font-size:11.5px;line-height:1.75">
      <div
style="padding:10px;background:rgba(125,211,199,.04);border-radius:6px"><strong
style="color:var(--seafoam)">Ocean Depth Slider:</strong> "Change the depth — watch the
speed update using  $v = \sqrt{gd}$ . Deeper ocean = faster tsunami."</div>
      <div
style="padding:10px;background:rgba(125,211,199,.04);border-radius:6px"><strong
style="color:var(--seafoam)">Wave Equation page:</strong> "Move frequency up — waves
compress. Move wavelength and speed changes. This is  $v = f\lambda$  live."</div>
      <div
style="padding:10px;background:rgba(125,211,199,.04);border-radius:6px"><strong
style="color:var(--seafoam)">5 Simulation Views:</strong> "Surface, Underground, Satellite,
Cross-Section, Dashboard — same event from different angles."</div>
      <div
style="padding:10px;background:rgba(125,211,199,.04);border-radius:6px"><strong
style="color:var(--seafoam)">Detection Log:</strong> "Start Earthquake — the log fills in real
time like a genuine mission control."</div>
      <div
style="padding:10px;background:rgba(245,158,11,.06);border-radius:6px"><strong
style="color:var(--amber)">Wave Science Hover Cards:</strong> "Hover any card — full
scientific explanation appears. All GCSE waves content covered."</div>
    </div>
  </div>
</div>
<!-- Q&A -->
<div class="omk-content" id="omk-c-3">
  <div class="g2">
    <div class="panel">
      <div class="sec-title" style="color:var(--red2)">Teacher Questions — Main</div>
      <div
style="display:flex;flex-direction:column;gap:10px;font-size:11.5px;line-height:1.7">
        <div style="padding:11px;background:rgba(59,130,246,.05);border-radius:6px"><div
style="color:var(--blue);font-weight:700;margin-bottom:2px">Q: Tsunami speed
formula?</div> $v = \sqrt{g \times d}$ . At 4000m:  $\sqrt{9.81 \times 4000} = 198 \text{ m/s} = 713 \text{ km/h}$ . At 10,000m:
 $\sqrt{9.81 \times 10000} = 313 \text{ m/s} = 1,127 \text{ km/h}$ .</div>
        <div style="padding:11px;background:rgba(59,130,246,.05);border-radius:6px"><div
style="color:var(--blue);font-weight:700;margin-bottom:2px">Q: Why do seismic waves travel
faster?</div>Rock is far more rigid and elastic than water. P-waves are longitudinal
compression waves through Earth's crust at 5–14 km/s. Water cannot support these.</div>
        <div style="padding:11px;background:rgba(59,130,246,.05);border-radius:6px"><div
style="color:var(--blue);font-weight:700;margin-bottom:2px">Q: P vs S

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waves?</div>P-waves: Primary, compressional, longitudinal, 14,000 km/h, through any medium. S-waves: Secondary, shear, transverse, 8,000 km/h, solid only.</div>

<div style="padding:11px;background:rgba(59,130,246,.05);border-radius:6px"><div style="color:var(--blue);font-weight:700;margin-bottom:2px">Q: How does a DART buoy work?</div>Bottom Pressure Recorder on seafloor measures water pressure anomalies. Passes via acoustic modem to surface buoy → satellite → warning centre.</div>

<div style="padding:11px;background:rgba(59,130,246,.05);border-radius:6px"><div style="color:var(--blue);font-weight:700;margin-bottom:2px">Q: Explain $v = f\lambda$?</div>Wave speed = frequency × wavelength. $f = 0.0005$ Hz, $\lambda = 200,000$ m → $v = 100$ m/s. The equation is universal for all waves.</div>

</div>

</div>

<div class="panel">

<div class="sec-title" style="color:var(--red2)">Advanced Questions</div>

<div

style="display:flex;flex-direction:column;gap:10px;font-size:11.5px;line-height:1.7">

<div style="padding:11px;background:rgba(245,158,11,.05);border-radius:6px"><div style="color:var(--amber);font-weight:700;margin-bottom:2px">Q: Coastal speed change?</div>As depth decreases: v drops ($v=\sqrt{gd}$). But energy is conserved, so amplitude grows. Shoaling: 0.5m deep ocean → 15m at coast. Wavelength also compresses — waves stack up.</div>

<div style="padding:11px;background:rgba(245,158,11,.05);border-radius:6px"><div style="color:var(--amber);font-weight:700;margin-bottom:2px">Q: How does hybrid reduce false alarms?</div>Only ~1 in 10 M6+ earthquakes generates a destructive tsunami. Seismic alone would alert constantly. Buoy confirmation acts as two-factor check — 85% fewer false alarms.</div>

<div style="padding:11px;background:rgba(245,158,11,.05);border-radius:6px"><div style="color:var(--amber);font-weight:700;margin-bottom:2px">Q: Is Wavara real?</div>Wavara is a fictional Pacific island nation representing real coastal communities at risk. All physics, speeds, and methods are based on NOAA DART II and Pacific Tsunami Warning Centre.</div>

<div style="padding:11px;background:rgba(245,158,11,.05);border-radius:6px"><div style="color:var(--amber);font-weight:700;margin-bottom:2px">Q: Future improvements?</div>AI pattern recognition, seafloor GPS (coseismic deformation), denser buoy networks, real-time ocean models for coastal run-up prediction.</div>

</div>

</div>

</div>

</div>

<!-- SUMMARY -->

<div class="omk-content" id="omk-c-4">

<div class="g2">

<div class="panel">

<div class="sec-title" style="color:var(--red2)">30-Second Summary</div>

<div style="display:flex;flex-direction:column;gap:12px">

<div style="padding:16px;background:rgba(125,211,199,.06);border:1px solid rgba(125,211,199,.15);border-radius:9px;font-size:14px;color:var(--text);line-height:1.8">"Operation WaveGuard is a hybrid tsunami early-warning system combining seismic P-wave

🔒 CLASSIFIED — OMK INTELLIGENCE — OPERATION WAVEGUARD —
EDUCATIONAL PURPOSE

```
</div>
</div>
</div>

<!-- APP -->
<div id="app" style="display:none">
  <nav id="navbar">
    <div style="display:flex;align-items:center;gap:8px;margin-right:14px;flex-shrink:0">
      <svg width="22" height="22" viewBox="0 0 52 52" fill="none"
style="color:var(--seafoam2);animation:insigniaPulse 5s ease-in-out infinite">
        <circle cx="26" cy="26" r="24" stroke="currentColor" stroke-width="1.3" opacity=".65"/>
        <path d="M4,26 C10,20 18,32 26,26 C34,20 42,32 48,26" stroke="currentColor"
stroke-width="2" stroke-linecap="round"/>
        <line x1="18" y1="33" x2="34" y2="33" stroke="currentColor" stroke-width="1"
stroke-linecap="round" opacity=".45"/>
        <line x1="21" y1="38" x2="31" y2="38" stroke="currentColor" stroke-width=".9"
stroke-linecap="round" opacity=".3"/>
      </svg>
      <span
style="color:var(--seafoam);font-size:10px;font-weight:800;letter-spacing:2.5px;text-transfor
m:uppercase;white-space:nowrap">WaveGuard</span>
      <div class="badge lvl-sci" style="font-size:7px">Monitoring</div>
    </div>
    <button class="nav-link active" onclick="showPage('home')">Home</button>
    <button class="nav-link" onclick="showPage('simulation')">Simulation</button>
    <button class="nav-link" onclick="showPage('wave')">Wave Science</button>
    <button class="nav-link" onclick="showPage('equation')"> $v = f\lambda$ </button>
    <button class="nav-link" onclick="showPage('comparison')">Comparison</button>
    <button class="nav-link" onclick="showPage('graphs')">Graphs</button>
    <button class="nav-link" onclick="showPage('scientists')">Scientists</button>
    <button class="nav-link" onclick="showPage('safety')">Safety</button>
    <div style="flex:1"></div>
    <div style="display:flex;align-items:center;gap:7px;margin-right:10px;flex-shrink:0">
      <div style="display:flex;align-items:center;gap:3px"><div
style="width:5px;height:5px;border-radius:50%;background:var(--green);"></div><span
class="mono"
style="font-size:7px;color:var(--muted2);text-transform:uppercase;letter-spacing:1px">Online
</span></div>
      <div style="display:flex;align-items:center;gap:3px"><div id="nav-sim-dot"
style="width:5px;height:5px;border-radius:50%;background:var(--border2)"></div><span
class="mono"
style="font-size:7px;color:var(--muted2);text-transform:uppercase;letter-spacing:1px">Sim</s
pan></div>
    </div>
    <button onclick="openOmkLock()" style="background:transparent;border:1px solid
rgba(36,52,71,.5);color:rgba(158,179,200,.2);padding:4px
```

```
10px;border-radius:4px;font-size:7.5px;letter-spacing:1.5px;text-transform:uppercase;cursor:
pointer;flex-shrink:0" title="OMK Classified">OMK</button>
</nav>
```

```
<!-- HOME -->
<div id="page-home" class="page active">
  <div style="min-height:calc(100vh - 43px);position:relative;overflow:hidden;background:
radial-gradient(
ellipse at top,
rgba(59,130,246,.16) 0%,
transparent 45%
),
```

```
linear-gradient(
180deg,
#081018 0%,
#111827 45%,
#0f172a 100%
)">
```

```
<div id="hero-stars" style="position:absolute;inset:0;pointer-events:none"></div>
```

```
<svg style="position:absolute;bottom:0;left:0;width:100%" viewBox="0 0 1440 300"
preserveAspectRatio="none" height="280">
```

```
<defs><linearGradient id="hg" x1="0" y1="0" x2="0" y2="1"><stop offset="0%"
stop-color="#0e2e4a" stop-opacity=".55"/><stop offset="100%" stop-color="#040c14"
stop-opacity="1"/></linearGradient></defs>
```

```
<path fill="url(#hg)" d="M0,112 C360,68 720,175 1080,112 L1440,112 L1440,300
L0,300 Z" style="animation:waveDrift 7s ease-in-out infinite"/>
```

```
<path fill="url(#hg)" fill-opacity=".4" d="M0,142 C440,95 880,205 1320,142 L1440,155
L1440,300 L0,300 Z" style="animation:waveDrift 9s ease-in-out infinite reverse"/>
```

```
<path fill="rgba(125,211,199,.035)" d="M0,172 C500,126 1000,224 1440,172
L1440,300 L0,300 Z"/>
```

```
</svg>
```

```
<div
```

```
style="position:relative;z-index:10;display:flex;flex-direction:column;align-items:center;justify-
content:center;min-height:calc(100vh - 43px);padding:40px 24px;text-align:center">
```

```
<div style="animation:fadeInUp .9s ease">
```

```
<div style="display:flex;justify-content:center;margin-bottom:18px">
```

```
<svg width="58" height="58" viewBox="0 0 52 52" fill="none"
```

```
style="color:var(--seafoam2);filter:drop-shadow(0 10px 30px
rgba(0,0,0,.35));animation:insigniaPulse 5s ease-in-out infinite">
```

```
<circle cx="26" cy="26" r="24" stroke="currentColor" stroke-width="1.2"
opacity=".65"/>
```

```
<circle cx="26" cy="26" r="14" stroke="currentColor" stroke-width=".6"
opacity=".3"/>
```

```
<path d="M4,26 C10,20 18,32 26,26 C34,20 42,32 48,26" stroke="currentColor"
stroke-width="2" stroke-linecap="round"/>
```

```

        <line x1="18" y1="33" x2="34" y2="33" stroke="currentColor" stroke-width=".9"
stroke-linecap="round" opacity=".45"/>
        <line x1="21" y1="38" x2="31" y2="38" stroke="currentColor" stroke-width=".8"
stroke-linecap="round" opacity=".3"/>
        <line x1="24" y1="43" x2="28" y2="43" stroke="currentColor" stroke-width=".7"
stroke-linecap="round" opacity=".18"/>
    </svg>
</div>
<div class="mono"
style="font-size:8px;letter-spacing:5px;color:var(--muted2);font-weight:600;text-transform:up
percase;margin-bottom:9px">WaveGuard Hybrid Warning System</div>
    <h1 style="font-family:'Instrument Serif',serif;
font-size:clamp(48px,8vw,110px);
font-weight:400;
letter-spacing:-0.05em;
line-height:.92;
color:#e5eef5;
text-transform:none;background:linear-gradient(135deg,var(--seafoam),#d8f5f2
50%,var(--seafoam2));-webkit-background-clip:text;-webkit-text-fill-color:transparent;line-heig
ht:.98;margin-bottom:8px">Operation<br/>WaveGuard</h1>
    <div class="mono"
style="font-size:11px;letter-spacing:4px;color:var(--muted2);text-transform:uppercase;margin
-bottom:8px">Hybrid Tsunami Early Warning System</div>
    <div
style="display:flex;justify-content:center;gap:7px;flex-wrap:wrap;margin-bottom:26px">
        <span class="badge lvl-1">Level 1 Monitoring</span>
        <span class="badge lvl-3">Level 3 Response</span>
        <span class="badge lvl-sci">Scientific Command</span>
    </div>
    <p style="font-size:13px;color:var(--muted);max-width:580px;margin:0 auto
32px;line-height:1.85">A dual-sensor hybrid system combining <span
style="color:var(--seafoam)">seismic P-wave detection</span> with <span
style="color:var(--blue)">DART ocean pressure buoy confirmation</span>, providing
maximum warning time with minimum false alarms for coastal populations.</p>
    <div style="display:flex;gap:10px;justify-content:center;flex-wrap:wrap">
        <button class="btn-p" onclick="showPage('simulation')" style="padding:12px
30px;font-size:11px">▶ Launch Simulation</button>
        <button class="tab-btn" onclick="showPage('wave')" style="padding:11px
20px;font-size:9px">~~ Wave Science</button>
        <button class="tab-btn" onclick="showPage('scientists')" style="padding:11px
20px;font-size:9px">👤 Scientists</button>
    </div>
</div>
<!-- Stats -->
<div
style="display:grid;grid-template-columns:repeat(4,1fr);gap:12px;margin-top:54px;max-width:
880px;width:100%;animation:fadeInUp 1.1s ease">
    <div class="glass-light" style="padding:16px 12px;text-align:center">

```

```

        <div class="mono"
style="font-size:8px;letter-spacing:2px;color:var(--muted2);text-transform:uppercase;margin-
bottom:5px">Epicentre → Wavara</div>
        <div class="mono" style="color:var(--seafoam);font-size:21px;font-weight:700">7,000
km</div>
        <div class="mono"
style="font-size:8px;color:rgba(158,179,200,.4);margin-top:3px">Pacific scale</div>
        </div>
        <div class="glass-light" style="padding:16px 12px;text-align:center">
        <div class="mono"
style="font-size:8px;letter-spacing:2px;color:var(--muted2);text-transform:uppercase;margin-
bottom:5px">Seismic Speed</div>
        <div class="mono" style="color:var(--amber);font-size:21px;font-weight:700">14,000
km/h</div>
        <div class="mono"
style="font-size:8px;color:rgba(158,179,200,.4);margin-top:3px">P-wave through crust</div>
        </div>
        <div class="glass-light" style="padding:16px 12px;text-align:center">
        <div class="mono"
style="font-size:8px;letter-spacing:2px;color:var(--muted2);text-transform:uppercase;margin-
bottom:5px">Tsunami Speed</div>
        <div class="mono" style="color:var(--blue);font-size:21px;font-weight:700">1,000
km/h</div>
        <div class="mono"
style="font-size:8px;color:rgba(158,179,200,.4);margin-top:3px">Deep ocean surface</div>
        </div>
        <div class="glass-light" style="padding:16px 12px;text-align:center">
        <div class="mono"
style="font-size:8px;letter-spacing:2px;color:var(--muted2);text-transform:uppercase;margin-
bottom:5px">Hybrid Advantage</div>
        <div class="mono" style="color:var(--green);font-size:21px;font-weight:700">6.5
hrs</div>
        <div class="mono"
style="font-size:8px;color:rgba(158,179,200,.4);margin-top:3px">vs. buoy-only</div>
        </div>
        </div>
        </div>
        </div>
        </div>
        </div>
        </div>

<!-- SIMULATION -->
<div id="page-simulation" class="page">
    <div class="wrap">
        <div class="sec-head"><span style="font-size:20px"><img alt="wave icon" data-bbox="588 828 608 843"/></span><h2>Hybrid System
Simulation</h2></div>
        <div style="display:flex;gap:7px;margin-bottom:11px;flex-wrap:wrap;align-items:center">
            <button class="tab-btn active" id="vbtn-surface"
onclick="setView('surface')">Surface</button>

```

```

        <button class="tab-btn" id="vbtn-underground"
onclick="setView('underground')">Underground</button>
        <button class="tab-btn" id="vbtn-satellite"
onclick="setView('satellite')">Satellite</button>
        <button class="tab-btn" id="vbtn-cross"
onclick="setView('cross')">Cross-Section</button>
        <button class="tab-btn" id="vbtn-dashboard"
onclick="setView('dashboard')">Dashboard</button>
        <div style="flex:1"></div>
        <button class="btn-d" id="sim-btn" onclick="startSim()"> ⚡ Start Earthquake</button>
        <button class="tab-btn" onclick="resetSim()" style="padding:6px 12px">↺
Reset</button>
    </div>
    <div id="sim-grid" style="display:grid;grid-template-columns:1fr
300px;gap:12px;margin-bottom:12px">
        <div id="sim-canvas">
            <!-- SURFACE -->
            <div id="view-surface" class="panel"
style="height:390px;padding:0;overflow:hidden;position:relative">
                <div style="position:absolute;inset:0;background:linear-gradient(180deg,#050e18
0%,#071a2e 28%,#082240 55%,#040c18 100%)"></div>
                <div style="position:absolute;inset:0;opacity:.1" class="gg"></div>
                <div
style="position:absolute;top:122px;left:0;right:0;height:1px;background:linear-gradient(90deg
,transparent,rgba(125,211,199,.22),rgba(59,130,246,.28),rgba(125,211,199,.22),transparent)
"></div>
                <div
style="position:absolute;top:123px;left:0;right:0;height:18px;opacity:.3;background:linear-gra
dient(90deg,transparent,rgba(49,92,103,.28) 25%,rgba(125,211,199,.25)
50%,rgba(49,92,103,.28) 75%,transparent);background-size:300px
100%;animation:waveScroll 3.5s linear infinite"></div>
                <div
style="position:absolute;top:142px;left:14px;font-size:7.5px;color:rgba(125,211,199,.28);lette
r-spacing:2px;text-transform:uppercase;font-family:'Courier New',monospace">Ocean
Surface</div>
                <div
style="position:absolute;bottom:62px;left:14px;font-size:7.5px;color:rgba(36,52,71,.8);letter-s
pacing:2px;text-transform:uppercase;font-family:'Courier New',monospace">Seafloor</div>
                <!-- Epicentre -->
                <div style="position:absolute;left:15%;bottom:66px;transform:translateX(-50%)>
                    <svg width="34" height="36" viewBox="0 0 34 36"><polygon points="17,3 32,33
2,33" fill="none" stroke="var(--red2)" stroke-width="1.4"/><text x="17" y="24"
text-anchor="middle" fill="var(--red2)" font-size="13"
font-family="monospace">!</text></svg>
                    <div id="surf-pulses"></div>
                    <div
style="font-size:6.5px;color:var(--red2);text-align:center;font-weight:700;letter-spacing:1px;m
argin-top:-1px;font-family:'Courier New',monospace">M8.2 EQ</div>

```

```

    </div>
    <!-- Seismic wave -->
    <div id="sv-swave"
style="position:absolute;bottom:60px;left:15%;display:none;transform:translateX(-50%)">
    <div
style="width:20px;height:20px;border-radius:50%;background:radial-gradient(circle,rgba(245
,158,11,.65),transparent);border:1.5px solid var(--amber);box-shadow:0 0 12px
var(--amber)"></div>
    <div
style="font-size:6.5px;color:var(--amber);text-align:center;margin-top:3px;font-weight:700;lett
er-spacing:1px;font-family:'Courier New',monospace">P-WAVE</div>
    </div>
    <!-- Tsunami -->
    <div id="sv-tsunami"
style="position:absolute;top:96px;left:15%;display:none;transform:translateX(-50%)">
    <svg width="50" height="52" viewBox="0 0 50 52"><path d="M0,44 Q12,8 25,22
Q38,36 50,12 L50,52 L0,52 Z" fill="rgba(59,130,246,.28)" stroke="var(--blue)"
stroke-width="1.4"/></svg>
    <div
style="font-size:6.5px;color:var(--blue);text-align:center;font-weight:700;letter-spacing:1px;fo
nt-family:'Courier New',monospace">TSUNAMI</div>
    </div>
    <!-- Sensors -->
    <div id="sv-s1"
style="position:absolute;left:33%;bottom:66px;transform:translateX(-50%)"><div
class="sv-sensor"
style="width:15px;height:15px;border-radius:2px;background:var(--panel2);border:1.4px
solid
var(--muted2);display:flex;align-items:center;justify-content:center;font-size:7.5px;color:var(--
muted2);transition:all .4s">▲</div><div
style="font-size:6.5px;text-align:center;color:var(--muted2);margin-top:2px;font-family:'Courie
r New',monospace">S1</div></div>
    <div id="sv-s2"
style="position:absolute;left:46%;bottom:66px;transform:translateX(-50%)"><div
class="sv-sensor"
style="width:15px;height:15px;border-radius:2px;background:var(--panel2);border:1.4px
solid
var(--muted2);display:flex;align-items:center;justify-content:center;font-size:7.5px;color:var(--
muted2);transition:all .4s">▲</div><div
style="font-size:6.5px;text-align:center;color:var(--muted2);margin-top:2px;font-family:'Courie
r New',monospace">S2</div></div>
    <!-- Buoys -->
    <div id="sv-b1"
style="position:absolute;left:43%;top:108px;transform:translateX(-50%);animation:buoyBob
3s ease-in-out infinite"><svg width="20" height="28" viewBox="0 0 20 28"><ellipse cx="10"
cy="8" rx="6.5" ry="4" fill="rgba(27,42,58,.8)" stroke="var(--muted2)" stroke-width="1.3"
class="buoy-top"/><line x1="10" y1="12" x2="10" y2="24" stroke="var(--muted2)"
stroke-width=".9"/><circle cx="10" cy="24" r="2" fill="var(--border2)"

```



```

class="buoy-base"/></svg><div
style="font-size:6.5px;text-align:center;color:var(--muted2);font-family:'Courier
New',monospace" class="buoy-lbl">B-1</div></div>
<div id="sv-b2"
style="position:absolute;left:58%;top:108px;transform:translateX(-50%);animation:buoyBob
3s ease-in-out infinite;animation-delay:.65s"><svg width="20" height="28" viewBox="0 0 20
28"><ellipse cx="10" cy="8" rx="6.5" ry="4" fill="rgba(27,42,58,.8)" stroke="var(--muted2)"
stroke-width="1.3" class="buoy-top"/><line x1="10" y1="12" x2="10" y2="24"
stroke="var(--muted2)" stroke-width=".9"/><circle cx="10" cy="24" r="2" fill="var(--border2)"
class="buoy-base"/></svg><div
style="font-size:6.5px;text-align:center;color:var(--muted2);font-family:'Courier
New',monospace" class="buoy-lbl">B-2</div></div>
<!-- Island -->
<div id="sv-island" style="position:absolute;right:4%;top:62px"><svg width="74"
height="92" viewBox="0 0 74 92"><ellipse cx="37" cy="74" rx="34" ry="15" fill="#1c3d1c"
stroke="#2a5a2a" stroke-width="1"/><ellipse cx="37" cy="60" rx="26" ry="20" fill="#214a21"
stroke="#2a5a2a" stroke-width="1"/><polygon points="28,42 37,10 46,42" fill="#2a6a2a"
stroke="#3a7a3a" stroke-width="1"/><rect x="34" y="32" width="7" height="12"
fill="#7a4a20" opacity=".8"/><rect x="17" y="58" width="6" height="8" fill="#1e3040"
class="sv-building"/><rect x="45" y="54" width="6" height="8" fill="#1e3040"
class="sv-building"/><rect x="31" y="61" width="6" height="8" fill="#1e3040"
class="sv-building"/></svg><div
style="font-size:7.5px;color:var(--green);text-align:center;letter-spacing:2px;font-weight:700;f
ont-family:'Courier New',monospace">WAVARA</div><div id="sv-island-alert"
style="display:none;font-size:7.5px;color:var(--red2);text-align:center;font-weight:700;animati
on:blinkAlert .7s step-end infinite;font-family:'Courier New',monospace">△
ALERT</div></div>
<!-- Control centre -->
<div style="position:absolute;right:4%;bottom:12px"><div id="sv-ctrl"
style="width:60px;height:44px;border-radius:5px;background:var(--panel);border:1.4px solid
var(--border);display:flex;flex-direction:column;align-items:center;justify-content:center;transiti
on:all .4s"><div style="font-size:14px">🖥️</div><div
style="font-size:6.5px;color:var(--muted2);letter-spacing:1px;font-family:'Courier
New',monospace" id="sv-ctrl-lbl">CTRL</div></div><svg width="22" height="22" viewBox="0
0 22 22" style="margin:3px auto 0;display:block"><circle cx="11" cy="11" r="10" fill="none"
stroke="rgba(125,211,199,.1)" stroke-width="1"/><circle cx="11" cy="11" r="2"
fill="var(--seafoam2)"><line x1="11" y1="11" x2="11" y2="2" stroke="rgba(125,211,199,.45)"
stroke-width="1" style="transform-origin:11px 11px;animation:radarSweep 3s linear
infinite"/></svg></div>
<div id="sv-arrows"
style="display:none;position:absolute;top:84px;left:20%;font-size:9px;color:var(--blue);font-fa
mily:'Courier New',monospace">→ Propagation</div>
<div
style="position:absolute;top:9px;left:13px;font-size:8.5px;font-weight:700;letter-spacing:2px;c
olor:rgba(125,211,199,.55);text-transform:uppercase;font-family:'Courier
New',monospace">Surface View</div>
<div id="sv-pbar"
style="position:absolute;bottom:0;left:0;right:0;height:2px;background:var(--border);display:n

```

```

one"><div id="sv-pfill" class="prog-fill"
style="background:linear-gradient(90deg,var(--accent),var(--seafoam))"></div></div>
    <div id="sv-complete"
style="display:none;position:absolute;top:0;left:0;right:0;background:rgba(16,185,129,.1);bor
der-bottom:1px solid
rgba(16,185,129,.35);padding:7px;text-align:center;font-size:10.5px;font-weight:700;color:var
(--green);letter-spacing:2px;text-transform:uppercase;font-family:'Courier
New',monospace">✓ Simulation Complete — WaveGuard Issued 6.5 Hour
Advantage</div>
    </div>
    <!-- UNDERGROUND -->
    <div id="view-underground" style="display:none" class="panel"
style="height:390px;padding:0;overflow:hidden;position:relative">
        <div
style="position:absolute;inset:0;height:390px;background:linear-gradient(180deg,#050e18
0%,#071830 42%,#3a1800 80%,#1e0d00 100%)"></div>
            <div style="position:absolute;inset:0;height:390px;opacity:.12" class="gg"></div>
            <div
style="position:absolute;bottom:0;left:0;right:0;height:95px;background:linear-gradient(180de
g,transparent,rgba(58,24,0,.75));border-top:1px dashed rgba(36,52,71,.45)"><div
style="position:absolute;top:7px;left:13px;font-size:7.5px;color:rgba(158,179,200,.3);letter-sp
acing:2px;text-transform:uppercase;font-family:'Courier New',monospace">Earth's Crust —
Oceanic Plate</div></div>
                <div style="position:absolute;left:15%;bottom:93px;transform:translateX(-50%)"><svg
width="32" height="34" viewBox="0 0 32 34"><polygon points="16,3 30,31 2,31" fill="none"
stroke="var(--red2)" stroke-width="1.3"/><text x="16" y="23" text-anchor="middle"
fill="var(--red2)" font-size="12" font-family="monospace">!</text></svg><div
id="ug-pulses"></div></div>
                    <svg id="ug-rings-svg"
style="position:absolute;left:15%;bottom:93px;transform:translateX(-50%)" width="1"
height="1" viewBox="-120 -120 240 240" overflow="visible">
                        <circle id="ugr1" cx="0" cy="0" r="0" fill="none" stroke="var(--amber)"
stroke-width="1.8" opacity=".65"/>
                        <circle id="ugr2" cx="0" cy="0" r="0" fill="none" stroke="var(--amber)"
stroke-width="1.3" opacity=".42"/>
                        <circle id="ugr3" cx="0" cy="0" r="0" fill="none" stroke="var(--amber)"
stroke-width=".9" opacity=".25"/>
                    </svg>
                    <div id="ug-swave"
style="position:absolute;bottom:88px;left:15%;display:none;transform:translateX(-50%)"><di
v
style="width:22px;height:22px;border-radius:50%;background:radial-gradient(circle,rgba(245
,158,11,.7),transparent);border:1.8px solid var(--amber);box-shadow:0 0 14px
var(--amber)"></div><div
style="font-size:6.5px;color:var(--amber);text-align:center;margin-top:3px;font-weight:700;lett
er-spacing:1px;font-family:'Courier New',monospace">P-WAVE<br/><span
style="color:var(--muted2)">14000 km/h</span></div></div>

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    <div style="position:absolute;left:33%;bottom:97px;transform:translateX(-50%)"><div
class="ug-sensor"
style="width:14px;height:14px;border-radius:2px;background:var(--panel2);border:1.3px
solid
var(--muted2);display:flex;align-items:center;justify-content:center;font-size:7px;color:var(--m
uted2);transition:all .4s">▲</div><div
style="font-size:6.5px;text-align:center;color:var(--muted2);margin-top:1px;font-family:'Courie
r New',monospace">S1</div></div>
    <div style="position:absolute;left:46%;bottom:97px;transform:translateX(-50%)"><div
class="ug-sensor"
style="width:14px;height:14px;border-radius:2px;background:var(--panel2);border:1.3px
solid
var(--muted2);display:flex;align-items:center;justify-content:center;font-size:7px;color:var(--m
uted2);transition:all .4s">▲</div><div
style="font-size:6.5px;text-align:center;color:var(--muted2);margin-top:1px;font-family:'Courie
r New',monospace">S2</div></div>
    <div
style="position:absolute;top:9px;left:13px;font-size:8.5px;font-weight:700;letter-spacing:2px;c
olor:rgba(245,158,11,.55);text-transform:uppercase;font-family:'Courier
New',monospace">Underground — Seismic P-Wave</div>
    <div id="ug-pbar"
style="position:absolute;bottom:0;left:0;right:0;height:2px;background:var(--border);display:n
one"><div id="ug-pfill" class="prog-fill"
style="background:linear-gradient(90deg,var(--amber),var(--amber2))"></div></div>
    </div>
    <!-- SATELLITE -->
    <div id="view-satellite" style="display:none" class="panel"
style="height:390px;padding:0;overflow:hidden;position:relative">
    <div style="position:absolute;inset:0;height:390px;background:radial-gradient(ellipse
at 50% 50%,#0c1e34 0%,#040c14 100%)"></div>
    <svg width="100%" height="390" viewBox="0 0 610 390"
style="position:absolute;inset:0">
    <g opacity=".07" stroke="var(--seafoam2)" stroke-width=".4">
    <line x1="100" y1="0" x2="100" y2="390"/><line x1="200" y1="0" x2="200"
y2="390"/>
    <line x1="300" y1="0" x2="300" y2="390"/><line x1="400" y1="0" x2="400"
y2="390"/>
    <line x1="500" y1="0" x2="500" y2="390"/>
    <line x1="0" y1="78" x2="610" y2="78"/><line x1="0" y1="156" x2="610"
y2="156"/>
    <line x1="0" y1="234" x2="610" y2="234"/><line x1="0" y1="312" x2="610"
y2="312"/>
    </g>
    <circle cx="98" cy="195" r="7.5" fill="var(--red2)" fill-opacity=".88"/>
    <circle id="sat-r1" cx="98" cy="195" r="0" fill="none" stroke="var(--red2)"
stroke-width="1.8" style="animation:pulseRing 2s ease-out infinite"/>
    <circle id="sat-r2" cx="98" cy="195" r="0" fill="none" stroke="var(--red2)"
stroke-width="1.4" style="animation:pulseRing 2s ease-out infinite;animation-delay:.65s"/>

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    <circle id="sat-tsunami-ring" cx="98" cy="195" r="0" fill="none" stroke="var(--blue)"
stroke-width="2" stroke-opacity=".75"/>
    <circle cx="265" cy="186" r="5.5" fill="var(--seafoam2)" fill-opacity=".65"
stroke="var(--seafoam)" stroke-width="1" id="sat-b1"/>
    <circle cx="352" cy="205" r="5.5" fill="var(--seafoam2)" fill-opacity=".65"
stroke="var(--seafoam)" stroke-width="1" id="sat-b2"/>
    <ellipse cx="482" cy="195" rx="40" ry="27" fill="#1e3a1e" stroke="#2a5a2a"
stroke-width="1.4"/>
    <text id="sat-warn" x="482" y="200" text-anchor="middle" fill="var(--red2)"
font-size="13" font-family="monospace" style="display:none;animation:blinkAlert .8s
step-end infinite">⚠</text>
    <text x="78" y="214" fill="rgba(239,68,68,.55)" font-size="8.5"
font-family="monospace">EQ</text>
    <text x="250" y="176" fill="var(--seafoam2)" font-size="8.5"
font-family="monospace">B1</text>
    <text x="338" y="196" fill="var(--seafoam2)" font-size="8.5"
font-family="monospace">B2</text>
    <text x="460" y="234" fill="var(--green)" font-size="8.5"
font-family="monospace">WAVARA</text>
    <line x1="18" y1="370" x2="116" y2="370" stroke="rgba(158,179,200,.25)"
stroke-width="1"/>
    <text x="18" y="367" fill="rgba(158,179,200,.35)" font-size="7"
font-family="monospace">0</text>
    <text x="96" y="367" fill="rgba(158,179,200,.35)" font-size="7"
font-family="monospace">3500km</text>
</svg>
<div
style="position:absolute;top:9px;left:13px;font-size:8.5px;font-weight:700;letter-spacing:2px;color:rgba(125,211,199,.55);text-transform:uppercase;font-family:'Courier
New',monospace">Satellite View — Top Down</div>
</div>
<!-- CROSS-SECTION -->
<div id="view-cross" style="display:none" class="panel"
style="height:390px;padding:0;overflow:hidden;position:relative">
    <div
style="position:absolute;inset:0;height:390px;background:linear-gradient(180deg,#050e18
0%,#071a2e 30%,#09244a 55%,#3a1800 78%,#1e0d00 100%)"></div>
    <div style="position:absolute;inset:0;height:390px;opacity:.1" class="gg"></div>
    <div
style="position:absolute;top:130px;left:0;right:0;height:1px;background:linear-gradient(90deg
,transparent,rgba(125,211,199,.26),rgba(59,130,246,.26),transparent)"></div>
    <div style="position:absolute;bottom:86px;left:0;right:0;height:1.5px;border-top:1px
dashed rgba(36,52,71,.55)"></div>
    <div
style="position:absolute;bottom:90px;left:13px;font-size:7px;color:rgba(158,179,200,.28);lett
er-spacing:2px;text-transform:uppercase;font-family:'Courier
New',monospace">Seafloor</div>

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    <div
style="position:absolute;top:134px;left:13px;font-size:7px;color:rgba(125,211,199,.28);letter-
spacing:2px;text-transform:uppercase;font-family:'Courier New',monospace">Ocean</div>
    <div style="position:absolute;left:15%;bottom:88px;transform:translateX(-50%)"><svg
width="30" height="32" viewBox="0 0 30 32"><polygon points="15,3 28,29 2,29" fill="none"
stroke="var(--red2)" stroke-width="1.3"/><text x="15" y="22" text-anchor="middle"
fill="var(--red2)" font-size="11" font-family="monospace">!</text></svg><div
id="cross-pulses"></div></div>
    <div id="cs-swave"
style="position:absolute;bottom:82px;left:15%;display:none;transform:translateX(-50%)"><di
v
style="width:19px;height:19px;border-radius:50%;background:radial-gradient(circle,rgba(245
,158,11,.65),transparent);border:1.5px solid var(--amber);box-shadow:0 0 11px
var(--amber)"></div><div
style="font-size:6.5px;color:var(--amber);text-align:center;margin-top:2px;font-weight:700;font-
family:'Courier New',monospace">P-WAVE</div></div>
    <div id="cs-tsunami"
style="position:absolute;top:106px;left:15%;display:none;transform:translateX(-50%)"><svg
width="46" height="44" viewBox="0 0 46 44"><path d="M0,36 Q11,6 23,19 Q35,32 46,8
L46,44 L0,44 Z" fill="rgba(59,130,246,.28)" stroke="var(--blue)"
stroke-width="1.4"/></svg><div
style="font-size:6.5px;color:var(--blue);text-align:center;font-weight:700;font-family:'Courier
New',monospace">WAVE</div></div>
    <div style="position:absolute;left:33%;bottom:88px;transform:translateX(-50%)"><div
class="cs-sensor"
style="width:14px;height:14px;border-radius:2px;background:var(--panel2);border:1.3px
solid
var(--muted2);display:flex;align-items:center;justify-content:center;font-size:7px;color:var(--m
uted2);transition:all .4s">▲</div></div>
    <div style="position:absolute;left:46%;bottom:88px;transform:translateX(-50%)"><div
class="cs-sensor"
style="width:14px;height:14px;border-radius:2px;background:var(--panel2);border:1.3px
solid
var(--muted2);display:flex;align-items:center;justify-content:center;font-size:7px;color:var(--m
uted2);transition:all .4s">▲</div></div>
    <div
style="position:absolute;left:43%;top:114px;transform:translateX(-50%);animation:buoyBob
3s ease-in-out infinite"><svg width="17" height="23" viewBox="0 0 17 23"><ellipse cx="8.5"
cy="6.5" rx="5.5" ry="3.5" fill="rgba(27,42,58,.8)" stroke="var(--muted2)" stroke-width="1.2"
class="cs-buoy-top"/><line x1="8.5" y1="10" x2="8.5" y2="19" stroke="var(--muted2)"
stroke-width=".8"/><circle cx="8.5" cy="19" r="2" fill="var(--border2)"
class="cs-buoy-top"/></svg></div>
    <div
style="position:absolute;left:58%;top:114px;transform:translateX(-50%);animation:buoyBob
3s ease-in-out infinite;animation-delay:.65s"><svg width="17" height="23" viewBox="0 0 17
23"><ellipse cx="8.5" cy="6.5" rx="5.5" ry="3.5" fill="rgba(27,42,58,.8)"
stroke="var(--muted2)" stroke-width="1.2" class="cs-buoy-top"/><line x1="8.5" y1="10"

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x2="8.5" y2="19" stroke="var(--muted2)" stroke-width=".8"/><circle cx="8.5" cy="19" r="2"
fill="var(--border2)" class="cs-buoy-top"/></svg></div>
    <div style="position:absolute;right:5%;top:65px"><svg width="66" height="85"
viewBox="0 0 66 85"><ellipse cx="33" cy="69" rx="30" ry="14" fill="#1c3d1c"
stroke="#2a5a2a" stroke-width="1"/><ellipse cx="33" cy="55" rx="22" ry="17" fill="#214a21"
stroke="#2a5a2a" stroke-width="1"/><polygon points="25,38 33,8 41,38" fill="#2a6a2a"
stroke="#3a7a3a" stroke-width="1"/><rect x="29" y="28" width="6" height="11"
fill="#7a4a20" opacity=".8"/></svg><div
style="font-size:7.5px;color:var(--green);text-align:center;letter-spacing:2px;font-weight:700;font-family:'Courier New',monospace">WAVARA</div></div>
    <div
style="position:absolute;top:9px;left:13px;font-size:8.5px;font-weight:700;letter-spacing:2px;color:rgba(125,211,199,.55);text-transform:uppercase;font-family:'Courier
New',monospace">Cross-Section View</div>
    <div id="cs-pbar"
style="position:absolute;bottom:0;left:0;right:0;height:2px;background:var(--border);display:none"><div id="cs-pfill" class="prog-fill"
style="background:linear-gradient(90deg,var(--accent),var(--seafoam))"></div></div>
    </div>
    <!-- DASHBOARD -->
    <div id="view-dashboard" style="display:none" class="panel"
style="height:390px;padding:12px">
    <div style="display:grid;grid-template-columns:1fr 1fr;grid-template-rows:1fr
1fr;gap:9px;height:100%">
    <div id="db-s" style="border:1px solid
var(--border);border-radius:7px;padding:11px;background:var(--bg);display:flex;flex-direction:
column;align-items:center;justify-content:center;transition:all .4s">
    <div style="font-size:22px;margin-bottom:7px">📶</div><svg width="40"
height="40" viewBox="0 0 40 40"><circle cx="20" cy="20" r="18" fill="none"
stroke="rgba(125,211,199,.1)" stroke-width="1"/><circle cx="20" cy="20" r="2.5"
fill="var(--seafoam2)"/><line x1="20" y1="20" x2="20" y2="4" stroke="rgba(125,211,199,.4)"
stroke-width="1" style="transform-origin:20px 20px;animation:radarSweep 3s linear
infinite"/></svg>
    <div id="db-s-lbl" class="mono"
style="font-size:8px;font-weight:700;letter-spacing:2px;color:var(--muted2);margin-top:7px;text-transform:uppercase">Seismic Net</div>
    <div id="db-s-st" class="mono"
style="font-size:8px;color:var(--border2);margin-top:3px">○ Standby</div>
    </div>
    <div id="db-b" style="border:1px solid
var(--border);border-radius:7px;padding:11px;background:var(--bg);display:flex;flex-direction:
column;align-items:center;justify-content:center;transition:all .4s">
    <div style="font-size:22px;margin-bottom:7px">🌊</div><svg width="40"
height="40" viewBox="0 0 40 40"><circle cx="20" cy="20" r="18" fill="none"
stroke="rgba(59,130,246,.1)" stroke-width="1"/><circle cx="20" cy="20" r="2.5"
fill="var(--blue)"/><line x1="20" y1="20" x2="20" y2="4" stroke="rgba(59,130,246,.4)"
stroke-width="1" style="transform-origin:20px 20px;animation:radarSweep 4s linear infinite
reverse"/></svg>

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        <div id="db-b-lbl" class="mono"
style="font-size:8px;font-weight:700;letter-spacing:2px;color:var(--muted2);margin-top:7px;te
xt-transform:uppercase">Buoy Net</div>
        <div id="db-b-st" class="mono"
style="font-size:8px;color:var(--border2);margin-top:3px">○ Standby</div>
    </div>
    <div id="db-h" style="border:1px solid
var(--border);border-radius:7px;padding:11px;background:var(--bg);display:flex;flex-direction:
column;align-items:center;justify-content:center;transition:all .4s">
        <div style="font-size:22px;margin-bottom:7px">🧠</div><svg width="40"
height="40" viewBox="0 0 40 40"><circle cx="20" cy="20" r="18" fill="none"
stroke="rgba(245,158,11,.1)" stroke-width="1"/><circle cx="20" cy="20" r="2.5"
fill="var(--amber)"/><line x1="20" y1="20" x2="20" y2="4" stroke="rgba(245,158,11,.38)"
stroke-width="1" style="transform-origin:20px 20px;animation:radarSweep 2.5s linear
infinite"/></svg>
        <div id="db-h-lbl" class="mono"
style="font-size:8px;font-weight:700;letter-spacing:2px;color:var(--muted2);margin-top:7px;te
xt-transform:uppercase">Hybrid AI</div>
        <div id="db-h-st" class="mono"
style="font-size:8px;color:var(--border2);margin-top:3px">○ Standby</div>
    </div>
    <div id="db-a" style="border:1px solid
var(--border);border-radius:7px;padding:11px;background:var(--bg);display:flex;flex-direction:
column;align-items:center;justify-content:center;transition:all .4s">
        <div style="font-size:22px;margin-bottom:7px">⚠️</div><svg width="40"
height="40" viewBox="0 0 40 40"><circle cx="20" cy="20" r="18" fill="none"
stroke="rgba(220,38,38,.1)" stroke-width="1"/><circle cx="20" cy="20" r="2.5"
fill="var(--red2)"/><line x1="20" y1="20" x2="20" y2="4" stroke="rgba(220,38,38,.38)"
stroke-width="1" style="transform-origin:20px 20px;animation:radarSweep 3.5s linear infinite
reverse"/></svg>
        <div id="db-a-lbl" class="mono"
style="font-size:8px;font-weight:700;letter-spacing:2px;color:var(--muted2);margin-top:7px;te
xt-transform:uppercase">Alert Sys</div>
        <div id="db-a-st" class="mono"
style="font-size:8px;color:var(--border2);margin-top:3px">○ Standby</div>
    </div>
</div>
</div>
</div>
<!-- RIGHT PANEL -->
<div style="display:flex;flex-direction:column;gap:11px">
    <div class="panel">
        <div class="sec-title">Live Calculations</div>
        <div style="display:flex;flex-direction:column;gap:0">
            <div
style="display:flex;justify-content:space-between;align-items:center;padding:6px
0;border-bottom:1px solid var(--border)"><span class="mono"
style="font-size:9px;color:var(--muted2)">Seismic Detection</span><span class="mono"

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id="dr-seis" style="font-size:14px;color:var(--muted2)">—<span style="font-size:8px">
min</span></span></div>
    <div
style="display:flex;justify-content:space-between;align-items:center;padding:6px
0;border-bottom:1px solid var(--border)"><span class="mono"
style="font-size:9px;color:var(--muted2)">Buoy Confirm</span><span class="mono"
id="dr-buoy" style="font-size:14px;color:var(--muted2)">—<span style="font-size:8px">
min</span></span></div>
    <div
style="display:flex;justify-content:space-between;align-items:center;padding:6px
0;border-bottom:1px solid var(--border)"><span class="mono"
style="font-size:9px;color:var(--muted2)">Tsunami Arrival</span><span class="mono"
id="dr-arrival" style="font-size:14px;color:var(--muted2)">—<span style="font-size:8px">
hrs</span></span></div>
    <div
style="display:flex;justify-content:space-between;align-items:center;padding:6px
0;border-bottom:1px solid var(--border)"><span class="mono"
style="font-size:9px;color:var(--muted2)">Warning Advantage</span><span class="mono"
id="dr-adv" style="font-size:14px;color:var(--green)">—<span style="font-size:8px">
hrs</span></span></div>
    <div
style="display:flex;justify-content:space-between;align-items:center;padding:6px 0"><span
class="mono" style="font-size:9px;color:var(--muted2)">Reliability</span><span
class="mono" id="dr-rel" style="font-size:14px;color:var(--green)">—<span
style="font-size:8px">%</span></span></div>
    </div>
    <div style="margin-top:11px"><div
style="display:flex;justify-content:space-between;margin-bottom:4px"><span class="mono"
style="font-size:7.5px;font-weight:700;letter-spacing:2px;color:var(--seafoam);text-transform:
uppercase">Threat Level</span><span id="threat-lbl" class="mono"
style="font-size:7.5px;font-weight:700;letter-spacing:2px;color:var(--green);text-transform:up
percase">NOMINAL</span></div><div class="bar"><div class="bar-fill" id="threat-fill"
style="width:0%;background:var(--green)"></div></div></div>
    <div id="emerg-panel"
style="display:none;margin-top:11px;padding:9px;background:rgba(220,38,38,.07);border:1px
solid rgba(239,68,68,.25);border-radius:6px;text-align:center"><div class="mono"
style="color:var(--red2);font-size:12px;font-weight:800;letter-spacing:2px;animation:blinkAlert
.8s step-end infinite">TSUNAMI WARNING<br/>ISSUED — WAVARA</div><div
class="mono" style="font-size:8px;color:var(--text);margin-top:5px">Hybrid advantage:
<strong style="color:var(--green)" id="adv-display">6.5 hrs</strong></div></div>
    </div>
    <div class="panel">
    <div class="sec-title">Sensor Network</div>
    <div style="display:flex;flex-direction:column;gap:0">
    <div
style="display:flex;justify-content:space-between;align-items:center;padding:5px
0;border-bottom:1px solid var(--border)"><span class="mono"

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style="font-size:8.5px;color:var(--muted2)">SEISMIC-1</span><span class="s-status mono"
data-id="s1" style="font-size:8.5px;color:var(--border2)">○ Standby</span></div>
    <div
style="display:flex;justify-content:space-between;align-items:center;padding:5px
0;border-bottom:1px solid var(--border)"><span class="mono"
style="font-size:8.5px;color:var(--muted2)">SEISMIC-2</span><span class="s-status mono"
data-id="s2" style="font-size:8.5px;color:var(--border2)">○ Standby</span></div>
    <div
style="display:flex;justify-content:space-between;align-items:center;padding:5px
0;border-bottom:1px solid var(--border)"><span class="mono"
style="font-size:8.5px;color:var(--muted2)">BUOY-ALPHA</span><span class="s-status
mono" data-id="b1" style="font-size:8.5px;color:var(--border2)">○ Standby</span></div>
    <div
style="display:flex;justify-content:space-between;align-items:center;padding:5px
0;border-bottom:1px solid var(--border)"><span class="mono"
style="font-size:8.5px;color:var(--muted2)">BUOY-BETA</span><span class="s-status
mono" data-id="b2" style="font-size:8.5px;color:var(--border2)">○ Standby</span></div>
    <div
style="display:flex;justify-content:space-between;align-items:center;padding:5px 0"><span
class="mono" style="font-size:8.5px;color:var(--muted2)">WARNING SYS</span><span
class="s-status mono" data-id="warn" style="font-size:8.5px;color:var(--border2)">○
Standby</span></div>
    </div>
</div>
<div class="panel">
    <div class="sec-title">Event Log</div>
    <div id="event-log" class="mono"
style="font-size:8.5px;color:var(--muted2);line-height:1.9;max-height:130px;overflow-y:auto"
><div>System standby...</div></div>
    </div>
</div>
<!-- Timeline -->
<div class="panel" style="margin-bottom:13px">
    <div class="sec-title">Detection & Warning Timeline — 7,000 km</div>
    <div style="position:relative;padding:18px 0 8px">
        <div
style="position:relative;height:26px;background:rgba(36,52,71,.45);border-radius:4px;overflow:
hidden;margin-bottom:26px">
            <div
style="position:absolute;left:0;top:0;bottom:0;width:7.1%;background:linear-gradient(90deg,rg
gba(245,158,11,.65),rgba(245,158,11,.45));border-right:2px solid var(--amber)"></div>
            <div id="tl-tsunami-fill"
style="position:absolute;left:0;top:0;bottom:0;width:0;background:linear-gradient(90deg,rgba(
59,130,246,.22),rgba(59,130,246,.12));transition:width 8s linear"></div>
        </div>
        <div style="display:flex;justify-content:space-between;margin-top:-18px">

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        <div style="text-align:center;max-width:75px"><div class="mono"
style="color:var(--amber);font-weight:700;font-size:8.5px">T + 30 min</div><div
class="mono" style="font-size:7.5px;color:var(--muted2)">Seismic det.</div></div>
        <div style="text-align:center"><div class="mono"
style="color:var(--muted2);font-size:7.5px">← 6.5 hour evacuation window →</div><div
class="mono" style="color:var(--green);font-weight:700;font-size:11px">HYBRID
ADVANTAGE</div></div>
        <div style="text-align:center;max-width:75px"><div class="mono"
style="color:var(--blue);font-weight:700;font-size:8.5px">T + 7.0 hrs</div><div class="mono"
style="font-size:7.5px;color:var(--muted2)">Tsunami arrives</div></div>
    </div>
</div>
<!-- Controls -->
<div class="g3">
    <div class="panel">
        <div class="sec-title">Earthquake</div>
        <div style="margin-bottom:11px"><div
style="display:flex;justify-content:space-between;margin-bottom:3px"><span class="mono"
style="font-size:7.5px;font-weight:700;letter-spacing:1.5px;color:var(--muted2);text-transform
:uppercase">Magnitude</span><span class="mono"
style="font-size:11px;color:var(--seafoam)" id="v-mag">7.2 M</span></div><input
type="range" min="5" max="9" step="0.1" value="7.2" id="sl-mag"
oninput="updateParams()"/></div>
        <div style="margin-bottom:11px"><div
style="display:flex;justify-content:space-between;margin-bottom:3px"><span class="mono"
style="font-size:7.5px;font-weight:700;letter-spacing:1.5px;color:var(--muted2);text-transform
:uppercase">Distance (km)</span><span class="mono"
style="font-size:11px;color:var(--seafoam)" id="v-dist">7000 km</span></div><input
type="range" min="1000" max="12000" step="100" value="7000" id="sl-dist"
oninput="updateParams()"/></div>
        <div style="padding:9px;background:var(--bg);border-radius:5px;border:1px solid
var(--border)"><div class="mono" style="color:var(--seafoam2);font-size:11px">t =
d/v</div><div class="mono" id="f-eq-val"
style="font-size:8.5px;color:var(--muted2);margin-top:2px">t = 7000/14000 = 0.50 hrs = 30
min</div></div>
    </div>
<div class="panel">
    <div class="sec-title">Ocean</div>
    <div style="margin-bottom:11px"><div
style="display:flex;justify-content:space-between;margin-bottom:3px"><span class="mono"
style="font-size:7.5px;font-weight:700;letter-spacing:1.5px;color:var(--muted2);text-transform
:uppercase">Seismic Speed</span><span class="mono"
style="font-size:11px;color:var(--amber)" id="v-sspd">14000 km/h</span></div><input
type="range" min="5000" max="25000" step="500" value="14000" id="sl-sspd"
oninput="updateParams()"/></div>
    <div style="margin-bottom:11px"><div
style="display:flex;justify-content:space-between;margin-bottom:3px"><span class="mono"

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style="font-size:7.5px;font-weight:700;letter-spacing:1.5px;color:var(--muted2);text-transform
:uppercase">Tsunami Speed</span><span class="mono"
style="font-size:11px;color:var(--blue)" id="v-tspd">1000 km/h</span></div><input
type="range" min="200" max="1200" step="25" value="1000" id="sl-tspd"
oninput="updateParams()"/></div>
    <div style="padding:9px;background:var(--bg);border-radius:5px;border:1px solid
var(--border)"><div class="mono" style="color:var(--seafoam2);font-size:11px">v =
√(g·d)</div><div class="mono" id="f-ts-val"
style="font-size:8.5px;color:var(--muted2);margin-top:2px">√(9.81×10200) ≈ 316 m/s ≈ 1000
km/h</div></div>
    </div>
    <div class="panel">
        <div class="sec-title">Sensors</div>
        <div style="margin-bottom:11px"><div
style="display:flex;justify-content:space-between;margin-bottom:3px"><span class="mono"
style="font-size:7.5px;font-weight:700;letter-spacing:1.5px;color:var(--muted2);text-transform
:uppercase">Ocean Depth</span><span class="mono"
style="font-size:11px;color:var(--seafoam)" id="v-depth">4000 m</span></div><input
type="range" min="500" max="12000" step="100" value="4000" id="sl-depth"
oninput="updateParams()"/></div>
            <div style="margin-bottom:11px"><div
style="display:flex;justify-content:space-between;margin-bottom:3px"><span class="mono"
style="font-size:7.5px;font-weight:700;letter-spacing:1.5px;color:var(--muted2);text-transform
:uppercase">Buoy Sensitivity</span><span class="mono"
style="font-size:11px;color:var(--blue)" id="v-bsens">80%</span></div><input type="range"
min="10" max="100" step="1" value="80" id="sl-bsens" oninput="updateParams()"/></div>
                <div style="padding:9px;background:var(--bg);border-radius:5px;border:1px solid
var(--border)"><div class="mono" id="timing-lines"
style="font-size:8.5px;color:var(--text);line-height:1.85">T+30 min: Seismic alert<br/>T+7.0
hrs: Tsunami arrives</div></div>
                    </div>
                </div>
            </div>
        </div>
    </div>

<!-- WAVE SCIENCE -->
<div id="page-wave" class="page">
    <div class="wrap">
        <div class="sec-head"><span style="font-size:20px">☞</span><h2>Wave
Science</h2></div>
        <div class="panel" style="margin-bottom:14px">
            <div class="sec-title">Interactive Wave Diagram</div>
            <div style="display:flex;gap:14px;margin-bottom:12px;flex-wrap:wrap">
                <div style="flex:1;min-width:190px"><div
style="display:flex;justify-content:space-between;margin-bottom:3px"><span class="mono"
style="font-size:7.5px;font-weight:700;letter-spacing:1.5px;color:var(--muted2);text-transform
:uppercase">Frequency</span><span class="mono"

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style="font-size:11px;color:var(--amber)" id="ws-fv">1 Hz</span></div><input type="range"
min="0.5" max="4" step="0.1" value="1" id="ws-f" oninput="updateWaveScience()"/></div>
<div style="flex:1;min-width:190px"><div
style="display:flex;justify-content:space-between;margin-bottom:3px"><span class="mono"
style="font-size:7.5px;font-weight:700;letter-spacing:1.5px;color:var(--muted2);text-transform
:uppercase">Amplitude</span><span class="mono" style="font-size:11px;color:var(--red2)"
id="ws-av">30</span></div><input type="range" min="10" max="44" step="1" value="30"
id="ws-a" oninput="updateWaveScience()"/></div>
</div>
<svg id="wave-sci-svg" width="100%" viewBox="0 0 560 100" style="overflow:visible">
<line x1="0" y1="50" x2="560" y2="50" stroke="var(--border2)" stroke-width="1"/>
<path id="ws-path" d="" fill="none" stroke="var(--seafoam2)" stroke-width="2.2"/>
<line id="ws-wl-line" x1="0" y1="62" x2="140" y2="62" stroke="var(--seafoam)"
stroke-width="1.3"/>
<text id="ws-wl-txt" x="70" y="73" text-anchor="middle" fill="var(--seafoam)"
font-size="9.5" font-family="'Courier New',monospace"> $\lambda$ </text>
<line x1="7" y1="50" x2="7" y2="20" stroke="var(--red2)" stroke-width="1.3"
stroke-dasharray="3 2"/>
<text x="15" y="37" fill="var(--red2)" font-size="9.5" font-family="'Courier
New',monospace">A</text>
<circle id="ws-crest" cx="140" cy="20" r="4" fill="var(--seafoam)"
style="filter:drop-shadow(0 0 4px var(--seafoam))"/>
</svg>
<div class="mono"
style="color:var(--muted2);font-size:9.5px;text-align:center;margin-top:5px"
id="ws-formula"> $v = f\lambda \cdot f = 1 \text{ Hz} \cdot T = 1.00 \text{ s}$ </div>
</div>
<div class="g4" style="margin-bottom:14px">
<div class="panel concept-c" onmouseenter="showC(this,'Amplitude','Maximum
displacement from equilibrium. Deep-ocean tsunami amplitude = 0.3–1m (invisible to ships).
Shoaling near coast amplifies to 30–40m. Energy conserved:  $E \propto A^2$ .)"
onmouseleave="hideC(this)" style="cursor:pointer;transition:all .3s"><div class="mono"
style="font-size:8px;font-weight:700;letter-spacing:1.5px;color:var(--red2);margin-bottom:5px
;text-transform:uppercase">Amplitude</div><div class="cc-body mono"
style="font-size:8.5px;color:var(--muted2)">Hover →</div></div>
<div class="panel concept-c" onmouseenter="showC(this,'Wavelength ( $\lambda$ )','Distance
between consecutive crests. Tsunami  $\lambda = 200$ –500 km. Wind waves  $\lambda \approx 100$ m. Long  $\lambda$ 
means entire water column moves when seabed rises — catastrophic energy.)"
onmouseleave="hideC(this)" style="cursor:pointer;transition:all .3s"><div class="mono"
style="font-size:8px;font-weight:700;letter-spacing:1.5px;color:var(--seafoam);margin-bottom
:5px;text-transform:uppercase">Wavelength  $\lambda$ </div><div class="cc-body mono"
style="font-size:8.5px;color:var(--muted2)">Hover →</div></div>
<div class="panel concept-c" onmouseenter="showC(this,'Frequency (f)','Waves per
second.  $f = 1/T$ . Tsunami:  $\sim 0.00003 \text{ Hz}$  (one wave every 10–60 min). Higher  $f =$  shorter  $\lambda$  at
constant speed ( $v=f\lambda$ ). Measured in Hertz (Hz).)" onmouseleave="hideC(this)"
style="cursor:pointer;transition:all .3s"><div class="mono"
style="font-size:8px;font-weight:700;letter-spacing:1.5px;color:var(--amber);margin-bottom:5

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px;text-transform:uppercase">Frequency f</div><div class="cc-body mono" style="font-size:8.5px;color:var(--muted2)">Hover →</div></div>

<div class="panel concept-c" onmouseenter="showC(this,'Time Period (T)','Time for one full wave cycle. T = 1/f. Tsunami: 10–60 minutes. Ocean swell: 10–20 seconds. Wind wave: 1–5 seconds. Longer period = more water displaced = more destructive.')" onmouseleave="hideC(this)" style="cursor:pointer;transition:all .3s"><div class="mono" style="font-size:8px;font-weight:700;letter-spacing:1.5px;color:var(--green);margin-bottom:5px;text-transform:uppercase">Period T</div><div class="cc-body mono" style="font-size:8.5px;color:var(--muted2)">Hover →</div></div>

<div class="panel concept-c" onmouseenter="showC(this,'Transverse Waves','Particles oscillate PERPENDICULAR to travel direction. Examples: light, water surface waves, seismic S-waves. Can travel through solids, liquids, and vacuums (EM waves only).')" onmouseleave="hideC(this)" style="cursor:pointer;transition:all .3s"><div class="mono" style="font-size:8px;font-weight:700;letter-spacing:1.5px;color:var(--seafoam2);margin-bottom:5px;text-transform:uppercase">Transverse</div><div class="cc-body mono" style="font-size:8.5px;color:var(--muted2)">Hover →</div></div>

<div class="panel concept-c" onmouseenter="showC(this,'Longitudinal Waves','Particles oscillate PARALLEL to travel direction (compression/rarefaction). Examples: sound, seismic P-waves. Faster because rock is rigid in compression. Cannot travel through liquids.')" onmouseleave="hideC(this)" style="cursor:pointer;transition:all .3s"><div class="mono" style="font-size:8px;font-weight:700;letter-spacing:1.5px;color:var(--amber);margin-bottom:5px;text-transform:uppercase">Longitudinal</div><div class="cc-body mono" style="font-size:8.5px;color:var(--muted2)">Hover →</div></div>

<div class="panel concept-c" onmouseenter="showC(this,'Energy Transfer','Waves transfer energy, not matter. Water particles orbit in small circles as wave passes — buoys bob in place. The energy travels at 1000 km/h; the water barely moves horizontally.')" onmouseleave="hideC(this)" style="cursor:pointer;transition:all .3s"><div class="mono" style="font-size:8px;font-weight:700;letter-spacing:1.5px;color:#a78bfa;margin-bottom:5px;text-transform:uppercase">Energy Transfer</div><div class="cc-body mono" style="font-size:8.5px;color:var(--muted2)">Hover →</div></div>

<div class="panel concept-c" onmouseenter="showC(this,'Wave Speed','v = fλ (all waves). v = √(gd) (tsunamis). At 4000m: 198 m/s = 712 km/h. At 10,200m: 316 m/s ≈ 1000 km/h. Seismic P-wave: 14,000 km/h — 14× faster.')" onmouseleave="hideC(this)" style="cursor:pointer;transition:all .3s"><div class="mono" style="font-size:8px;font-weight:700;letter-spacing:1.5px;color:var(--blue);margin-bottom:5px;text-transform:uppercase">Wave Speed</div><div class="cc-body mono" style="font-size:8.5px;color:var(--muted2)">Hover →</div></div>

</div>

<div class="g2">

<div class="panel"><div class="sec-title" style="color:var(--seafoam2)">Transverse Wave — S-wave / Ocean Surface</div><svg id="tv-svg" width="100%" viewBox="0 0 520 96"><line x1="0" y1="48" x2="520" y2="48" stroke="var(--border2)" stroke-width="1" stroke-dasharray="4 3"/><path id="tv-path" d="" fill="none" stroke="var(--seafoam2)" stroke-width="1.9"/><text x="9" y="13" fill="var(--muted2)" font-size="8.5" font-family="Courier New,monospace">▶ Direction of travel</text><text x="9" y="91" fill="var(--seafoam2)" font-size="8.5" font-family="Courier New,monospace">↑ Particle motion — Perpendicular — Light, Surface Waves, S-waves</text></svg></div>

```

    <div class="panel"><div class="sec-title" style="color:var(--amber)">Longitudinal Wave
    — P-wave / Sound</div><svg id="lv-svg" width="100%" viewBox="0 0 520 96"><line x1="0"
    y1="48" x2="520" y2="48" stroke="var(--border2)" stroke-width="1" stroke-dasharray="4
    3"/><g id="lv-dots"></g><text x="9" y="13" fill="var(--muted2)" font-size="8.5"
    font-family="Courier New',monospace">> Direction of travel</text><text x="9" y="91"
    fill="var(--amber)" font-size="8.5" font-family="Courier New',monospace"><-> Particle motion
    — Parallel — Sound, P-waves, Compression waves</text></svg></div>
  </div>
</div>
</div>

```

```

<!-- EQUATION -->
<div id="page-equation" class="page">
  <div class="wrap">
    <div class="sec-head"><span style="font-size:20px">📐 </span><h2>Wave Equation —
    v = fλ</h2></div>
    <div class="panel">
      <div class="sec-title">Interactive Visualiser</div>
      <div
        style="display:flex;gap:22px;align-items:center;margin-bottom:18px;flex-wrap:wrap">
        <div style="padding:18px 30px;background:var(--bg);border-radius:9px;border:1px
        solid var(--border);text-align:center;flex-shrink:0"><div
        style="font-size:42px;font-weight:900;letter-spacing:2px;background:linear-gradient(135deg,
        var(--seafoam),#d8f5f2,var(--seafoam2));-webkit-background-clip:text;-webkit-text-fill-color:tr
        ansparent">v = fλ</div></div>
        <div style="flex:1;min-width:200px">
          <div style="margin-bottom:11px"><div
            style="display:flex;justify-content:space-between;margin-bottom:3px"><span class="mono"
            style="font-size:7.5px;font-weight:700;letter-spacing:1.5px;color:var(--muted2);text-transform
            :uppercase">Frequency (f)</span><span class="mono"
            style="font-size:11px;color:var(--amber)" id="eq-fv">2 Hz</span></div><input type="range"
            min="0.5" max="5" step="0.1" value="2" id="eq-f" oninput="updateEq()"/></div>
          <div><div
            style="display:flex;justify-content:space-between;margin-bottom:3px"><span class="mono"
            style="font-size:7.5px;font-weight:700;letter-spacing:1.5px;color:var(--muted2);text-transform
            :uppercase">Wavelength (λ)</span><span class="mono"
            style="font-size:11px;color:var(--seafoam)" id="eq-wv">80 m</span></div><input
            type="range" min="20" max="200" step="5" value="80" id="eq-w"
            oninput="updateEq()"/></div>
          </div>
          <div style="display:flex;flex-direction:column;gap:8px">
            <div style="padding:8px 14px;background:var(--bg);border-radius:5px;border:1px
            solid rgba(245,158,11,.18);min-width:130px"><div class="mono"
            style="font-size:7.5px;color:var(--muted2);letter-spacing:1px">Frequency</div><div
            class="mono" id="eq-f-disp" style="color:var(--amber);font-size:15px;font-weight:700">2
            Hz</div></div>
            <div style="padding:8px 14px;background:var(--bg);border-radius:5px;border:1px
            solid rgba(125,211,199,.18);min-width:130px"><div class="mono"

```

```
style="font-size:7.5px;color:var(--muted2);letter-spacing:1px">Wavelength</div><div
class="mono" id="eq-w-disp" style="color:var(--seafoam);font-size:15px;font-weight:700">80
m</div></div>
```

```
    <div style="padding:8px 14px;background:var(--bg);border-radius:5px;border:1px
solid rgba(16,185,129,.18);min-width:130px"><div class="mono"
style="font-size:7.5px;color:var(--muted2);letter-spacing:1px">Wave Speed</div><div
class="mono" id="eq-spdp-disp" style="color:var(--green);font-size:15px;font-weight:700">160
m/s</div></div>
```

```
    </div>
```

```
  </div>
```

```
    <svg id="eq-svg" width="100%" viewBox="0 0 560 115"><line x1="0" y1="57" x2="560"
y2="57" stroke="var(--border2)" stroke-width="1"/><path id="eq-path" d="" fill="none"
stroke="var(--seafoam2)" stroke-width="2.2" style="filter:drop-shadow(0 0 4px
rgba(91,192,190,.35))"/><line id="eq-wl-line" x1="20" y1="104" x2="100" y2="104"
stroke="var(--seafoam)" stroke-width="1.3"/><text id="eq-wl-ann" x="60" y="113"
text-anchor="middle" fill="var(--seafoam)" font-size="9.5" font-family="'Courier
New',monospace">λ = 80m</text></svg>
```

```
    <div
```

```
style="margin-top:11px;padding:11px;background:var(--bg);border-radius:5px;border:1px
solid var(--border)"><div class="mono" id="eq-formula"
style="color:var(--text);font-size:11px">v = f × λ = 2 × 80 = 160 m/s · T = 1/f = 0.500
s</div></div>
```

```
    </div>
```

```
  </div>
```

```
</div>
```

```
<!-- COMPARISON -->
```

```
<div id="page-comparison" class="page">
```

```
  <div class="wrap">
```

```
    <div class="sec-head"><span style="font-size:20px">⚖️</span><h2>System
Comparison</h2></div>
```

```
    <div style="display:flex;gap:9px;margin-bottom:16px;flex-wrap:wrap">
```

```
      <button class="tab-btn active" id="cmp-s-btn" onclick="selectSys('seismic')"
style="flex:1;padding:10px">System A — Seismic Only</button>
```

```
      <button class="tab-btn" id="cmp-b-btn" onclick="selectSys('buoy')"
style="flex:1;padding:10px">System B — Buoy Only</button>
```

```
      <button class="tab-btn" id="cmp-h-btn" onclick="selectSys('hybrid')"
style="flex:1;padding:10px">★ Hybrid WaveGuard</button>
```

```
    </div>
```

```
    <div class="g2">
```

```
      <div class="panel"><div class="sec-title" id="cmp-title"
style="color:var(--seafoam)">System A — Seismic Only</div><p id="cmp-desc"
style="font-size:11.5px;color:var(--text);line-height:1.75;margin-bottom:14px">Seismic
sensors detect earthquakes instantly but cannot confirm if a tsunami was generated. High
speed but elevated false alarm risk.</p><div id="cmp-metrics"></div></div>
```

```
      <div class="panel">
```

```
        <div class="sec-title">Performance Radar — All Systems</div>
```

```
        <svg id="radar-svg" width="100%" viewBox="0 0 300 280">
```

```

<polygon id="rp2" fill="none" stroke="var(--border)" stroke-width=".7"/>
<polygon id="rp4" fill="none" stroke="var(--border)" stroke-width=".7"/>
<polygon id="rp6" fill="none" stroke="var(--border)" stroke-width=".7"/>
<polygon id="rp8" fill="none" stroke="var(--border)" stroke-width=".7"/>
<polygon id="rp10" fill="none" stroke="var(--border)" stroke-width=".7"/>
<text x="150" y="11" text-anchor="middle" fill="var(--muted2)" font-size="7.5"
font-family="Courier New',monospace">Detection Speed</text>
<text x="278" y="82" fill="var(--muted2)" font-size="7.5" font-family="Courier
New',monospace">Reliability</text>
<text x="232" y="228" text-anchor="middle" fill="var(--muted2)" font-size="7.5"
font-family="Courier New',monospace">Evacuation</text>
<text x="68" y="228" text-anchor="middle" fill="var(--muted2)" font-size="7.5"
font-family="Courier New',monospace">False Alarm↓</text>
<text x="10" y="82" fill="var(--muted2)" font-size="7.5" font-family="Courier
New',monospace">Low Cost↓</text>
<polygon id="radar-seismic" fill="rgba(125,211,199,.07)" stroke="var(--seafoam2)"
stroke-width="1.4"/>
<polygon id="radar-buoy" fill="rgba(59,130,246,.07)" stroke="var(--blue)"
stroke-width="1" stroke-opacity=".55"/>
<polygon id="radar-hybrid" fill="rgba(16,185,129,.18)" stroke="var(--green)"
stroke-width="2.3"/>
<rect x="9" y="232" width="8" height="8" fill="var(--seafoam2)"
fill-opacity=".65"/><text x="22" y="240" fill="var(--muted2)" font-size="8.5"
font-family="Courier New',monospace">Seismic Only</text>
<rect x="9" y="247" width="8" height="8" fill="var(--blue)" fill-opacity=".65"/><text
x="22" y="255" fill="var(--muted2)" font-size="8.5" font-family="Courier
New',monospace">Buoy Only</text>
<rect x="9" y="262" width="8" height="8" fill="var(--green)" fill-opacity=".65"/><text
x="22" y="270" fill="var(--green)" font-size="8.5" font-family="Courier New',monospace"
font-weight="bold">Hybrid WaveGuard ★</text>
</svg>
</div>
</div>
</div>
</div>

```

```

<!-- GRAPHS -->
<div id="page-graphs" class="page">
<div class="wrap">
<div class="sec-head"><span style="font-size:20px"><img alt="Bar chart icon" data-bbox="588 758 608 778"/></span><h2>Live Data
Graphs</h2></div>
<div class="g3" style="margin-bottom:14px">
<div class="panel"><div class="sec-title">Tsunami Speed vs Depth</div><svg
id="g-depth-svg" width="100%" viewBox="0 0 260 162" class="gg"><line x1="35" y1="11"
x2="35" y2="135" stroke="var(--border2)" stroke-width="1"/><line x1="35" y1="135"
x2="252" y2="135" stroke="var(--border2)" stroke-width="1"/><text x="143" y="155"
text-anchor="middle" fill="var(--muted2)" font-size="7.5" font-family="Courier
New',monospace">Depth (m)</text><text x="35" y="146" text-anchor="middle"

```


fill="var(--muted2)" font-size="6.5">500</text><text x="143" y="146" text-anchor="middle"
fill="var(--muted2)" font-size="6.5">4000</text><text x="252" y="146" text-anchor="middle"
fill="var(--muted2)" font-size="6.5">8000</text><path id="g-depth-path" d="" fill="none"
stroke="var(--seafoam2)" stroke-width="1.8" style="filter:drop-shadow(0 0 3px
rgba(91,192,190,.35))"/><line id="g-depth-vline" x1="143" y1="11" x2="143" y2="135"
stroke="var(--seafoam)" stroke-width="1" stroke-dasharray="3 2"/><circle id="g-depth-dot"
cx="143" cy="70" r="3.5" fill="var(--seafoam)"/></svg><div class="mono" id="g-depth-lbl"
style="font-size:9.5px;color:var(--seafoam);text-align:center;margin-top:4px">4000m → 712
km/h</div></div>

<div class="panel"><div class="sec-title">Warning Effectiveness</div><div
style="display:flex;flex-direction:column;gap:9px;margin-top:6px"><div><div
style="display:flex;justify-content:space-between;margin-bottom:2px"><span class="mono"
style="font-size:8.5px;color:var(--muted2)">Hybrid WaveGuard<span class="mono"
style="font-size:9.5px;color:var(--green)">95%</div><div class="bar"
style="height:9px"><div class="bar-fill"
style="width:95%;background:linear-gradient(90deg,rgba(16,185,129,.4),var(--green))"></div>
</div></div><div><div
style="display:flex;justify-content:space-between;margin-bottom:2px"><span class="mono"
style="font-size:8.5px;color:var(--muted2)">Seismic Only<span class="mono"
style="font-size:9.5px;color:var(--seafoam2)">72%</div><div class="bar"
style="height:9px"><div class="bar-fill"
style="width:72%;background:linear-gradient(90deg,rgba(91,192,190,.4),var(--seafoam2))">
</div></div></div><div><div
style="display:flex;justify-content:space-between;margin-bottom:2px"><span class="mono"
style="font-size:8.5px;color:var(--muted2)">Buoy Only<span class="mono"
style="font-size:9.5px;color:var(--blue)">58%</div><div class="bar"
style="height:9px"><div class="bar-fill"
style="width:58%;background:linear-gradient(90deg,rgba(59,130,246,.4),var(--blue))"></div>
</div></div><div><div
style="display:flex;justify-content:space-between;margin-bottom:2px"><span class="mono"
style="font-size:8.5px;color:var(--muted2)">No System<span class="mono"
style="font-size:9.5px;color:var(--muted2)">5%</div><div class="bar"
style="height:9px"><div class="bar-fill"
style="width:5%;background:var(--muted2)"></div></div></div></div><div class="mono"
style="margin-top:11px;font-size:8.5px;color:var(--muted2);text-align:center">Effective
evacuation rate</div></div>

<div class="panel"><div class="sec-title">Signal Speed Comparison</div><svg
id="g-speed-svg" width="100%" viewBox="0 0 260 162"><g id="g-speed-bars"></g><text
id="g-speed-note" x="9" y="155" fill="var(--muted2)" font-size="7.5" font-family="Courier
New',monospace">Seismic ~14× faster</text></svg></div>

</div>

<div class="panel">

<div class="sec-title">Adjust Parameters</div>

<div class="g3">

<div><div

style="display:flex;justify-content:space-between;margin-bottom:3px"><span class="mono"
style="font-size:7.5px;font-weight:700;letter-spacing:1.5px;color:var(--muted2);text-transform
:uppercase">Magnitude<span class="mono"

```

style="font-size:11px;color:var(--seafoam)" id="gr-mag-v">7.2 M</span></div><input
type="range" min="5" max="9" step="0.1" value="7.2" id="gr-mag"
oninput="syncGraph('mag',this.value)"/></div>
<div><div
style="display:flex;justify-content:space-between;margin-bottom:3px"><span class="mono"
style="font-size:7.5px;font-weight:700;letter-spacing:1.5px;color:var(--muted2);text-transform
:uppercase">Ocean Depth</span><span class="mono"
style="font-size:11px;color:var(--seafoam2)" id="gr-depth-v">4000 m</span></div><input
type="range" min="500" max="12000" step="100" value="4000" id="gr-depth"
oninput="syncGraph('depth',this.value)"/></div>
<div><div
style="display:flex;justify-content:space-between;margin-bottom:3px"><span class="mono"
style="font-size:7.5px;font-weight:700;letter-spacing:1.5px;color:var(--muted2);text-transform
:uppercase">Seismic Speed</span><span class="mono"
style="font-size:11px;color:var(--amber)" id="gr-sspd-v">14000 km/h</span></div><input
type="range" min="5000" max="25000" step="500" value="14000" id="gr-sspd"
oninput="syncGraph('sspd',this.value)"/></div>
</div>
</div>
</div>
</div>

```

```

<!-- SCIENTISTS -->
<div id="page-scientists" class="page">
<div class="wrap">
<div class="sec-head"><span style="font-size:20px">🔬</span><h2>Scientific
Command</h2></div>
<div style="padding:16px
22px;margin-bottom:20px;background:linear-gradient(135deg,rgba(49,92,103,.15),rgba(59,1
30,246,.08));border:1px solid
var(--border);border-radius:9px;display:flex;align-items:center;justify-content:space-between;
flex-wrap:wrap;gap:12px">
<div><div class="mono"
style="font-size:7.5px;letter-spacing:3px;color:var(--muted2);text-transform:uppercase;margi
n-bottom:3px">Operation WaveGuard</div><div
style="font-size:17px;font-weight:800;color:var(--seafoam);letter-spacing:2px;text-transform:
uppercase">Wavara Defense Scientific Command</div><div class="mono"
style="font-size:9.5px;color:var(--muted2);margin-top:2px">Pacific Tsunami Early Warning
Network</div></div>
<div style="display:flex;gap:7px;flex-wrap:wrap"><div class="badge lvl-5">Level 5
Clearance</div><div class="badge lvl-sci">Scientific Command</div></div>
</div>
<div class="g3" style="margin-bottom:20px">
<div class="sci-card" style="animation:fadeInUp .7s ease">
<div
style="position:absolute;top:0;left:0;right:0;height:2px;background:linear-gradient(90deg,tran
SPARENT, var(--amber),transparent)"/></div>
<div style="display:flex;align-items:center;gap:11px;margin-bottom:13px">

```

```

    <div
style="width:50px;height:50px;border-radius:50%;display:flex;align-items:center;justify-conte
nt:center;font-size:21px;border:1.5px solid
rgba(245,158,11,.35);background:linear-gradient(135deg,rgba(245,158,11,.1),var(--panel))">
👤 </div>
    <div><div
style="font-size:15px;font-weight:800;color:var(--text);letter-spacing:1px">Omar</div><div
class="mono"
style="font-size:8px;color:var(--amber);letter-spacing:2px;text-transform:uppercase;margin-to
p:1px">Lead Seismic Analyst</div></div>
    </div>
    <div style="display:flex;flex-direction:column;gap:6px;margin-bottom:13px">
    <div style="display:flex;justify-content:space-between;padding:5px
9px;background:var(--bg);border-radius:4px;border:1px solid var(--border)"><span
class="mono" style="font-size:8.5px;color:var(--muted2)">Specialisation</span><span
class="mono" style="font-size:8.5px;color:var(--amber);font-weight:600">P-Wave
Analysis</span></div>
    <div style="display:flex;justify-content:space-between;padding:5px
9px;background:var(--bg);border-radius:4px;border:1px solid var(--border)"><span
class="mono" style="font-size:8.5px;color:var(--muted2)">Department</span><span
class="mono" style="font-size:8.5px;color:var(--text)">Seismic Intelligence</span></div>
    </div>
    <div style="display:flex;gap:5px;flex-wrap:wrap;margin-bottom:11px"><div
class="badge lvl-5" style="font-size:7px">Level 5</div><div
style="display:flex;align-items:center;gap:3px;padding:3px
7px;background:rgba(245,158,11,.07);border:1px solid
rgba(245,158,11,.2);border-radius:3px"><div
style="width:4px;height:4px;border-radius:50%;background:var(--green);"></div><span
class="mono"
style="font-size:6.5px;color:var(--muted2);letter-spacing:1.5px;text-transform:uppercase">On
Duty</span></div></div>
    <div class="mono"
style="font-size:9.5px;color:var(--muted2);line-height:1.7;border-top:1px solid
var(--border);padding-top:10px">Monitors global seismic networks for P-wave signatures
indicating undersea earthquakes. Coordinates rapid threat assessment and preliminary alert
decisions.</div>
    </div>
    <div class="sci-card" style="animation:fadeInUp .85s ease">
    <div
style="position:absolute;top:0;left:0;right:0;height:2px;background:linear-gradient(90deg,tran
SPARENT, var(--blue),transparent)"></div>
    <div style="display:flex;align-items:center;gap:11px;margin-bottom:13px">
    <div
style="width:50px;height:50px;border-radius:50%;display:flex;align-items:center;justify-conte
nt:center;font-size:21px;border:1.5px solid
rgba(59,130,246,.35);background:linear-gradient(135deg,rgba(59,130,246,.1),var(--panel))">
👤 </div>

```

```

    <div><div
style="font-size:15px;font-weight:800;color:var(--text);letter-spacing:1px">Karma</div><div
class="mono"
style="font-size:8px;color:var(--blue);letter-spacing:2px;text-transform:uppercase;margin-top:
1px">Ocean Monitoring Specialist</div></div>
</div>
    <div style="display:flex;flex-direction:column;gap:6px;margin-bottom:13px">
    <div style="display:flex;justify-content:space-between;padding:5px
9px;background:var(--bg);border-radius:4px;border:1px solid var(--border)"><span
class="mono" style="font-size:8.5px;color:var(--muted2)">Specialisation</span><span
class="mono" style="font-size:8.5px;color:var(--blue);font-weight:600">DART Buoy
Network</span></div>
    <div style="display:flex;justify-content:space-between;padding:5px
9px;background:var(--bg);border-radius:4px;border:1px solid var(--border)"><span
class="mono" style="font-size:8.5px;color:var(--muted2)">Department</span><span
class="mono" style="font-size:8.5px;color:var(--text)">Ocean Intelligence</span></div>
</div>
    <div style="display:flex;gap:5px;flex-wrap:wrap;margin-bottom:11px"><div
class="badge lvl-5" style="font-size:7px">Level 5</div><div
style="display:flex;align-items:center;gap:3px;padding:3px
7px;background:rgba(59,130,246,.07);border:1px solid
rgba(59,130,246,.2);border-radius:3px"><div
style="width:4px;height:4px;border-radius:50%;background:var(--green);"></div><span
class="mono"
style="font-size:6.5px;color:var(--muted2);letter-spacing:1.5px;text-transform:uppercase">On
Duty</span></div></div>
    <div class="mono"
style="font-size:9.5px;color:var(--muted2);line-height:1.7;border-top:1px solid
var(--border);padding-top:10px">Manages the DART deep-ocean buoy network across the
Pacific Basin. Interprets pressure anomalies to confirm tsunami presence with 98.7%
accuracy before full alerts.</div>
</div>
    <div class="sci-card" style="animation:fadeInUp 1s ease">
    <div
style="position:absolute;top:0;left:0;right:0;height:2px;background:linear-gradient(90deg,tran
sparent,var(--green),transparent)"></div>
    <div style="display:flex;align-items:center;gap:11px;margin-bottom:13px">
    <div
style="width:50px;height:50px;border-radius:50%;display:flex;align-items:center;justify-conte
nt:center;font-size:21px;border:1.5px solid
rgba(16,185,129,.35);background:linear-gradient(135deg,rgba(16,185,129,.1),var(--panel))">
 </div>
    <div><div
style="font-size:15px;font-weight:800;color:var(--text);letter-spacing:1px">Mariam</div><div
class="mono"
style="font-size:8px;color:var(--green);letter-spacing:2px;text-transform:uppercase;margin-to
p:1px">Hybrid Systems Coordinator</div></div>
</div>

```

```

<div style="display:flex;flex-direction:column;gap:6px;margin-bottom:13px">
  <div style="display:flex;justify-content:space-between;padding:5px
9px;background:var(--bg);border-radius:4px;border:1px solid var(--border)"><span
class="mono" style="font-size:8.5px;color:var(--muted2)">Specialisation</span><span
class="mono" style="font-size:8.5px;color:var(--green);font-weight:600">AI Fusion
Systems</span></div>
  <div style="display:flex;justify-content:space-between;padding:5px
9px;background:var(--bg);border-radius:4px;border:1px solid var(--border)"><span
class="mono" style="font-size:8.5px;color:var(--muted2)">Department</span><span
class="mono" style="font-size:8.5px;color:var(--text)">Systems Command</span></div>
</div>
<div style="display:flex;gap:5px;flex-wrap:wrap;margin-bottom:11px"><div
class="badge lvl-sci" style="font-size:7px">Sci Command</div><div
style="display:flex;align-items:center;gap:3px;padding:3px
7px;background:rgba(16,185,129,.07);border:1px solid
rgba(16,185,129,.2);border-radius:3px"><div
style="width:4px;height:4px;border-radius:50%;background:var(--green);"></div><span
class="mono"
style="font-size:6.5px;color:var(--muted2);letter-spacing:1.5px;text-transform:uppercase">On
Duty</span></div></div>
<div class="mono"
style="font-size:9.5px;color:var(--muted2);line-height:1.7;border-top:1px solid
var(--border);padding-top:10px">Coordinates the hybrid AI decision system. Manages the
false-alarm reduction algorithm that increased evacuation compliance by 340%.</div>
</div>
</div>
<div class="panel">
  <div class="sec-title">Authority &amp; Clearance System</div>
  <div class="g4">
    <div style="padding:13px;background:var(--bg);border-radius:7px;border:1px solid
rgba(16,185,129,.18);text-align:center"><div class="badge lvl-1"
style="font-size:7px;margin:0 auto 7px">Level 1</div><div class="mono"
style="font-size:10px;font-weight:700;color:var(--green);margin-bottom:3px">Monitoring</div>
<div class="mono" style="font-size:8.5px;color:var(--muted2);line-height:1.6">Read-only
access. Sensor feeds. Shared displays only.</div></div>
    <div style="padding:13px;background:var(--bg);border-radius:7px;border:1px solid
rgba(59,130,246,.18);text-align:center"><div class="badge lvl-3"
style="font-size:7px;margin:0 auto 7px">Level 3</div><div class="mono"
style="font-size:10px;font-weight:700;color:var(--blue);margin-bottom:3px">Response</div>
<div class="mono" style="font-size:8.5px;color:var(--muted2);line-height:1.6">Operational
access. Trigger protocols. Supervised.</div></div>
    <div style="padding:13px;background:var(--bg);border-radius:7px;border:1px solid
rgba(245,158,11,.18);text-align:center"><div class="badge lvl-5"
style="font-size:7px;margin:0 auto 7px">Level 5</div><div class="mono"
style="font-size:10px;font-weight:700;color:var(--amber);margin-bottom:3px">Full
Clearance</div><div class="mono"
style="font-size:8.5px;color:var(--muted2);line-height:1.6">Full system access. National
warnings. Algorithm control.</div></div>
  </div>
</div>

```

```
<div style="padding:13px;background:var(--bg);border-radius:7px;border:1px solid
rgba(125,211,199,.18);text-align:center"><div class="badge lvl-sci"
style="font-size:7px;margin:0 auto 7px">Sci CMD</div><div class="mono"
style="font-size:10px;font-weight:700;color:var(--seafoam);margin-bottom:3px">Scientific
Command</div><div class="mono"
style="font-size:8.5px;color:var(--muted2);line-height:1.6">OMK access. Classified briefings.
Omar, Karma, Mariam.</div></div>
</div>
</div>
</div>
</div>
```

```
<!-- SAFETY -->
<div id="page-safety" class="page">
  <div class="wrap">
    <div class="sec-head"><span style="font-size:20px"> 🏃 </span><h2>Public
Safety</h2></div>
    <div style="padding:18px
22px;margin-bottom:18px;background:linear-gradient(135deg,rgba(220,38,38,.09),rgba(245,
158,11,.06));border:1px solid
rgba(220,38,38,.28);border-radius:9px;display:flex;align-items:center;gap:18px;flex-wrap:wra
p">
      <div style="font-size:42px;flex-shrink:0"> ⚠️ </div>
      <div><div
style="color:var(--red2);font-size:13px;font-weight:800;letter-spacing:2px;text-transform:uppe
rcase;margin-bottom:5px">Public Tsunami Safety Guide</div><div
style="font-size:13px;color:var(--text);max-width:620px;line-height:1.75">WaveGuard issues
warnings <strong style="color:var(--amber)">6.5 hours before tsunami arrival</strong>.
Every second counts — follow these steps immediately upon receiving an alert.</div></div>
      <div class="g3" style="margin-bottom:18px">
        <div class="panel" style="text-align:center"><div
style="font-size:30px;margin-bottom:7px"> 🚒 </div><div class="mono"
style="font-size:7.5px;font-weight:700;letter-spacing:2px;color:var(--seafoam);text-transform:
uppercase;margin-bottom:3px">Step 1</div><div
style="font-size:11.5px;font-weight:700;color:var(--text);margin-bottom:7px">Hear the
Alert</div><div class="mono"
style="font-size:9.5px;color:var(--muted2);line-height:1.65">Listen for official warnings via
radio, TV, sirens, or mobile alerts. Do not wait to see the wave.</div></div>
        <div class="panel" style="text-align:center"><div
style="font-size:30px;margin-bottom:7px"> 🏃 </div><div class="mono"
style="font-size:7.5px;font-weight:700;letter-spacing:2px;color:var(--seafoam);text-transform:
uppercase;margin-bottom:3px">Step 2</div><div
style="font-size:11.5px;font-weight:700;color:var(--text);margin-bottom:7px">Move
Immediately</div><div class="mono"
style="font-size:9.5px;color:var(--muted2);line-height:1.65">High ground or inland. 1+ miles
from coast or 30 metres above sea level. You have 6.5 hours.</div></div>
```

[illegible]


```
    </div>
  </div>
</div>
</div>
```

```
    <footer style="border-top:1px solid
var(--border);padding:14px;text-align:center;background:var(--panel)">
    <div
style="display:flex;align-items:center;justify-content:center;gap:9px;margin-bottom:3px">
      <svg width="12" height="12" viewBox="0 0 52 52" fill="none"
style="color:rgba(125,211,199,.3)"><circle cx="26" cy="26" r="24" stroke="currentColor"
stroke-width="1.3" opacity=".6"/><path d="M4,26 C10,20 18,32 26,26 C34,20 42,32 48,26"
stroke="currentColor" stroke-width="2" stroke-linecap="round"/></svg>
      <span class="mono"
style="font-size:7.5px;font-weight:700;letter-spacing:2.5px;color:rgba(158,179,200,.35);text-t
ransform:uppercase">Operation WaveGuard · Wavara Defense Network · Omar · Karma ·
Mariam</span>
    </div>
    <div class="mono" style="font-size:8.5px;color:rgba(36,52,71,.8)">Educational project.
Based on real tsunami physics, NOAA DART II technology, and Pacific Tsunami Warning
Centre methodology.</div>
  </footer>
</div>
```

```
<script>
/* ===== SECURITY CONSTANTS ===== */
var MAIN_PW = 'RIPT1DE';
var LVL5_PW = 'ABBYS-07';
var OMK_PW = 'POSIED0N';
var DENY_KEY = 'wg_denied';
var FAIL_KEY = 'wg_fails';
var LOCK_KEY = 'wg_locked';
var MAX_FAILS = 3;

/* ===== STARTUP — CHECK SECURITY STATE ===== */
(function() {
  var denied = localStorage.getItem(DENY_KEY) === '1';
  var locked = localStorage.getItem(LOCK_KEY) === '1';
  if (denied) {
    showDenied();
  } else if (locked) {
    showLockdown();
  } else {
    // Show normal lock screen (already visible by default)
    renderAttemptRow();
  }
})();
```



```

function renderAttemptRow() {
  var fails = parseInt(localStorage.getItem(FAIL_KEY) || '0');
  var row = document.getElementById('attempt-row');
  if (!row) return;
  row.innerHTML = "";
  for (var i = 0; i < MAX_FAILS; i++) {
    var d = document.createElement('div');
    d.style.cssText = 'width:8px;height:8px;border-radius:50%;border:1px solid ' + (i < fails ?
'rgba(239,68,68,.7)' : 'var(--border2)') + ';background:' + (i < fails ? 'var(--red2)' :
'transparent') + ';transition:all .3s';
    row.appendChild(d);
  }
  if (fails > 0) {
    var left = MAX_FAILS - fails;
    var info = document.createElement('div');
    info.style.cssText =
'font-size:7.5px;color:rgba(239,68,68,.6);letter-spacing:1.5px;font-family:\'Courier
New\',monospace;margin-left:8px;text-transform:uppercase';
    info.textContent = left + ' attempt' + (left !== 1 ? 's' : '') + ' remaining';
    row.appendChild(info);
  }
}

```

```

function checkLock() {
  var v = (document.getElementById('lock-input').value || '').trim().toUpperCase();
  if (v === MAIN_PW) {
    localStorage.removeItem(FAIL_KEY);
    localStorage.removeItem(LOCK_KEY);
    var p = document.getElementById('lock-panel');
    p.style.animation = 'lockReveal .5s ease forwards';
    setTimeout(function() {
      document.getElementById('lock-screen').style.display = 'none';
      var ls = document.getElementById('loading-screen');
      ls.style.display = 'flex';
      startLoad();
    }, 480);
  } else {
    var fails = parseInt(localStorage.getItem(FAIL_KEY) || '0') + 1;
    localStorage.setItem(FAIL_KEY, fails);
    renderAttemptRow();
    var e = document.getElementById('lock-error');
    var p = document.getElementById('lock-panel');
    if (fails >= MAX_FAILS) {
      localStorage.setItem(LOCK_KEY, '1');
      e.textContent = 'Maximum attempts exceeded — lockdown initiated';
      e.style.opacity = '1';
      p.style.animation = 'shake .5s ease';
      setTimeout(function() { p.style.animation = ''; }, 520);
    }
  }
}

```

```

        setTimeout(function() { showLockdown(); }, 1200);
    } else {
        e.textContent = 'Invalid access code — ' + (MAX_FAILS - fails) + ' attempt' +
(MAX_FAILS - fails !== 1 ? 's' : '') + ' remaining';
        e.style.opacity = '1';
        p.style.animation = 'shake .5s ease';
        setTimeout(function() { p.style.animation = ''; }, 520);
        setTimeout(function() { e.style.opacity = '0'; }, 3500);
    }
    document.getElementById('lock-input').value = '';
    document.getElementById('lock-input').focus();
}
}
document.getElementById('lock-input').addEventListener('keydown', function(e) { if (e.key
=== 'Enter') checkLock(); });

```

```

function showLockdown() {
    document.getElementById('lock-screen').style.display = 'none';
    document.getElementById('loading-screen').style.display = 'none';
    var ld = document.getElementById('lockdown-screen');
    ld.style.display = 'flex';
    ld.style.flexDirection = 'column';
    ld.style.alignItems = 'center';
    ld.style.justifyContent = 'center';
    document.getElementById('ld-input').value = '';
}

```

```

function checkLvl5() {
    var v = (document.getElementById('ld-input').value || '').trim().toUpperCase();
    if (v === LVL5_PW) {
        localStorage.removeItem(FAIL_KEY);
        localStorage.removeItem(LOCK_KEY);
        localStorage.removeItem(DENY_KEY);
        document.getElementById('lockdown-screen').style.display = 'none';
        var ls = document.getElementById('loading-screen');
        ls.style.display = 'flex';
        startLoad();
    } else {
        localStorage.setItem(DENY_KEY, '1');
        var e = document.getElementById('ld-error');
        e.textContent = 'Invalid Level 5 code — access permanently revoked';
        e.style.opacity = '1';
        var p = document.getElementById('ld-panel');
        p.style.animation = 'shake .5s ease';
        setTimeout(function() { p.style.animation = ''; }, 520);
        setTimeout(function() {
            document.getElementById('lockdown-screen').style.display = 'none';
            showDenied();
        }, 3500);
    }
}

```

```

    }, 1200);
  }
}
document.getElementById('ld-input').addEventListener('keydown', function(e) { if (e.key ===
'Enter') checkLvl5(); });

function showDenied() {
  document.getElementById('lock-screen').style.display = 'none';
  document.getElementById('loading-screen').style.display = 'none';
  document.getElementById('lockdown-screen').style.display = 'none';
  var ds = document.getElementById('denial-screen');
  ds.style.display = 'flex';
  ds.style.flexDirection = 'column';
  var ref = document.getElementById('denial-ref');
  if (ref) ref.textContent = 'WG-' + Date.now().toString(36).toUpperCase();
}

/* ===== OMK ===== */
function checkOmk() {
  var v = (document.getElementById('omk-input').value || "").trim().toUpperCase();
  if (v === OMK_PW) {
    document.getElementById('omk-lock-screen').style.display = 'none';
    document.getElementById('omk-overlay').style.display = 'block';
  } else {
    var e = document.getElementById('omk-error');
    e.textContent = 'Incorrect classified code';
    e.style.opacity = '1';
    var p = document.getElementById('omk-lock-panel');
    p.style.animation = 'shake .5s ease';
    setTimeout(function() { p.style.animation = ""; }, 520);
    setTimeout(function() { e.style.opacity = '0'; }, 3000);
    document.getElementById('omk-input').value = "";
  }
}
document.getElementById('omk-input').addEventListener('keydown', function(e) { if (e.key
=== 'Enter') checkOmk(); });
function openOmkLock() {
  var s = document.getElementById('omk-lock-screen');
  s.style.display = 'flex';
  s.style.alignItems = 'center';
  s.style.justifyContent = 'center';
  document.getElementById('omk-input').value = "";
}
function closeOmkLock() { document.getElementById('omk-lock-screen').style.display =
'none'; }
function closeOmk() { document.getElementById('omk-overlay').style.display = 'none'; }
function setOmkTab(i) {
  for (var j = 0; j < 5; j++) {

```

```

var t = document.getElementById('omk-t-' + j);
var c = document.getElementById('omk-c-' + j);
if (!t || !c) continue;
if (j === i) {
    t.classList.add('active');
    t.style.color = 'var(--red2)';
    t.style.borderColor = 'rgba(239,68,68,.35)';
    c.classList.add('active');
} else {
    t.classList.remove('active');
    t.style.color = 'var(--muted2)';
    t.style.borderColor = 'var(--border)';
    c.classList.remove('active');
}
}
}

/* ===== LOADING ===== */
var bootLines = [
    '> WAVARA DEFENSE NETWORK — BOOT SEQUENCE INITIATED',
    '> [ OK ] Seismic sensor array — 47 stations online',
    '> [ OK ] DART buoy network — 12 units active',
    '> [ OK ] Satellite uplink — signal locked',
    '> [ OK ] Ocean telemetry — depth readings stable',
    '> [ OK ] Hybrid AI fusion engine — calibrated',
    '> [ OK ] Warning broadcast system — armed',
    '> SYSTEM READY — ALL SENSORS NOMINAL',
];
var bootIndex = 0;
function startLoad() {
    var prog = 0;
    var bar = document.getElementById('load-bar');
    var ph = document.getElementById('load-phase');
    var pct = document.getElementById('load-pct');
    var log = document.getElementById('boot-log');
    var phases = ['Initialising network...', 'Loading sensor array...', 'Calibrating DART',
        'buoys...', 'Connecting satellite...', 'Synchronising telemetry...', 'System ready'];
    bootIndex = 0;
    if (log) log.innerHTML = '';
    var iv = setInterval(function() {
        prog += Math.random() * 3.2 + 0.8;
        if (prog >= 100) {
            prog = 100;
            clearInterval(iv);
            if (bar) bar.style.width = '100%';
            if (ph) ph.textContent = phases[5];
            if (pct) pct.textContent = '100%';
            setTimeout(function() {

```

```

        document.getElementById('loading-screen').style.display = 'none';
        document.getElementById('app').style.display = 'block';
        initApp();
    }, 500);
} else {
    if (bar) bar.style.width = prog + '%';
    var pi = Math.min(Math.floor((prog / 100) * phases.length), phases.length - 2);
    if (ph) ph.textContent = phases[pi];
    if (pct) pct.textContent = Math.round(prog) + '%';
    // Boot log
    var li = Math.floor((prog / 100) * bootLines.length);
    if (log && li > bootIndex && bootIndex < bootLines.length) {
        var d = document.createElement('div');
        d.style.animation = 'bootText .2s ease';
        d.textContent = bootLines[bootIndex];
        log.appendChild(d);
        log.scrollTop = log.scrollHeight;
        bootIndex++;
    }
}
}, 55);
}

/* ===== STARS ===== */
function genStars(id, n) {
    var c = document.getElementById(id);
    if (!c) return;
    for (var i = 0; i < n; i++) {
        var d = document.createElement('div');
        var sz = Math.random() * 1.6 + 0.4;
        d.style.cssText = 'position:absolute;left:' + Math.random() * 100 + '%;top:' + Math.random()
* 70 + '%;width:' + sz + 'px;height:' + sz + 'px;border-radius:50%;background:white;opacity:' +
(Math.random() * 0.45 + 0.12) + ';animation:starTwinkle ' + (Math.random() * 3 + 2) + 's
step-end infinite;animation-delay:' + (Math.random() * 4) + 's';
        c.appendChild(d);
    }
}
genStars('ls-stars', 50);

/* ===== STATE ===== */
var P = { magnitude:7.2, distance:7000, seismicSpeed:14000, tsunamiSpeed:1000,
oceanDepth:4000, buoySensitivity:80 };
var simState = 'idle', simRAF = null, simStart = null;
var sDetected = false, bDetected = false, wIssued = false;
var seisPos = 0, tsuPos = 0;
var currentView = 'surface';
var waveRAF = null, eqRAF = null;

```

```

/* ===== NAV ===== */
function showPage(id) {
    document.querySelectorAll('.page').forEach(function(p) { p.classList.remove('active'); });
    var pg = document.getElementById('page-' + id);
    if (pg) pg.classList.add('active');
    document.querySelectorAll('.nav-link').forEach(function(b) { b.classList.remove('active'); });
    var nb = document.querySelector('.nav-link[onclick*="' + id + '"]');
    if (nb) nb.classList.add('active');
    window.scrollTo({ top: 0, behavior: 'smooth' });
    if (id === 'wave') startWaveAnim(); else stopWaveAnim();
    if (id === 'equation') startEqAnim(); else stopEqAnim();
    if (id === 'comparison') { initRadar(); selectSys('seismic'); }
    if (id === 'graphs') updateGraphs();
    if (id === 'safety') document.getElementById('safety-date').textContent = new
Date().toUTCString();
}

/* ===== CALCULATIONS ===== */
function seisTime() { return (P.distance / P.seismicSpeed) * 60; }
function tsuTime() { return P.distance / P.tsunamiSpeed; }
function hybridAdv() { return Math.max(0, tsuTime() - seisTime() / 60); }
function reliab() { return Math.min(99.9, 70 + P.buoySensitivity * 0.25 + (P.magnitude > 6 ? 8
: 0)); }
function threatPct() { return Math.min(100, Math.max(0, ((P.magnitude - 5) / 4) * 100)); }
function setText(id, t) { var e = document.getElementById(id); if (e) e.textContent = t; }

function updateParams() {
    P.magnitude = parseFloat(document.getElementById('sl-mag').value);
    P.distance = parseFloat(document.getElementById('sl-dist').value);
    P.seismicSpeed = parseFloat(document.getElementById('sl-sspd').value);
    P.tsunamiSpeed = parseFloat(document.getElementById('sl-tspd').value);
    P.oceanDepth = parseFloat(document.getElementById('sl-depth').value);
    P.buoySensitivity = parseFloat(document.getElementById('sl-bsens').value);
    setText('v-mag', P.magnitude.toFixed(1) + ' M');
    setText('v-dist', P.distance + ' km');
    setText('v-sspd', P.seismicSpeed + ' km/h');
    setText('v-tspd', P.tsunamiSpeed + ' km/h');
    setText('v-depth', P.oceanDepth + ' m');
    setText('v-bsens', P.buoySensitivity + '%');
    var sm = seisTime().toFixed(1);
    setText('f-eq-val', 't = ' + P.distance + '/' + P.seismicSpeed + ' = ' + sm + ' min');
    var tc = Math.round(Math.sqrt(9.81 * P.oceanDepth) * 3.6);
    setText('f-ts-val', 'v =  $\sqrt{9.81 \times P.oceanDepth}$   $\approx$  ' + Math.round(Math.sqrt(9.81 *
P.oceanDepth)) + ' m/s  $\approx$  ' + tc + ' km/h');
    setText('timing-lines', 'T+' + sm + 'min: Seismic alert\nT+' + tsuTime().toFixed(1) + 'hrs:
Tsunami arrives');
    updateLiveData(); updateGraphs(); syncGrSliders();
}

```

```

function updateLiveData() {
    var sm = seisTime(), ta = tsuTime(), adv = hybridAdv(), rel = reliab(), thr = threatPct();
    setText('dr-seis', sm.toFixed(1));
    setText('dr-buoy', (sm * 3).toFixed(0));
    setText('dr-arrival', ta.toFixed(2));
    setText('dr-adv', adv.toFixed(2));
    setText('dr-rel', rel.toFixed(1));
    setText('adv-display', adv.toFixed(1) + ' hrs');
    var color = thr < 30 ? 'var(--green)' : thr < 65 ? 'var(--amber)' : 'var(--red2)';
    var lbl = thr < 20 ? 'NOMINAL' : thr < 40 ? 'ELEVATED' : thr < 65 ? 'GUARDED' : thr < 85 ?
    'HIGH' : 'CRITICAL';
    var tf = document.getElementById('threat-fill');
    if (tf) { tf.style.width = thr + '%'; tf.style.background = 'linear-gradient(90deg,var(--accent),' +
    color + ')'; }
    var tl = document.getElementById('threat-lbl');
    if (tl) { tl.textContent = lbl; tl.style.color = color; }
}

```

```

function syncGrSliders() {
    var gm = document.getElementById('gr-mag');
    if (!gm) return;
    gm.value = P.magnitude; document.getElementById('gr-depth').value = P.oceanDepth;
    document.getElementById('gr-sspd').value = P.seismicSpeed;
    setText('gr-mag-v', P.magnitude.toFixed(1) + ' M'); setText('gr-depth-v', P.oceanDepth + '
    m'); setText('gr-sspd-v', P.seismicSpeed + ' km/h');
}

```

```

function syncGraph(field, val) {
    if (field === 'mag') { P.magnitude = parseFloat(val); var s =
    document.getElementById('sl-mag'); if(s).value=val; setText('v-mag',
    parseFloat(val).toFixed(1)+' M'); setText('gr-mag-v', parseFloat(val).toFixed(1)+' M'); }
    if (field === 'depth') { P.oceanDepth = parseFloat(val); var s =
    document.getElementById('sl-depth'); if(s).value=val; setText('v-depth', val+' m');
    setText('gr-depth-v', val+' m'); var tc=Math.round(Math.sqrt(9.81*P.oceanDepth)*3.6);
    setText('f-ts-val','v =  $\sqrt{(9.81 \times P.oceanDepth)}$   $\approx$ 
    '+Math.round(Math.sqrt(9.81*P.oceanDepth))+' m/s  $\approx$  '+tc+' km/h'); }
    if (field === 'sspd') { P.seismicSpeed = parseFloat(val); var s =
    document.getElementById('sl-sspd'); if(s).value=val; setText('v-sspd', val+' km/h');
    setText('gr-sspd-v', val+' km/h'); }
    updateLiveData(); updateGraphs();
}

```

```

/* ===== SIMULATION ===== */

```

```

function setView(v) {
    currentView = v;
    ['surface', 'underground', 'satellite', 'cross', 'dashboard'].forEach(function(id) {
        var el = document.getElementById('view-' + id);
    });
}

```

```

    if (el) el.style.display = (id === v) ? 'block' : 'none';
  });
  document.querySelectorAll("[id^='vbtn-']").forEach(function(b) { b.classList.remove('active');
});
  var b = document.getElementById('vbtn-' + v);
  if (b) b.classList.add('active');
}

```

```

function startSim() {
  if (simState === 'running') return;
  resetSim();
  simState = 'running'; simStart = Date.now(); sDetected = bDetected = wIssued = false;
  seisPos = tsuPos = 0;
  document.getElementById('sim-btn').disabled = true;
  document.getElementById('sim-btn').textContent = '● RUNNING...';
  ['sv-pbar','ug-pbar','cs-pbar'].forEach(function(id){ var e=document.getElementById(id);
if(e)e.style.display='block'; });
  document.getElementById('sv-complete').style.display='none';
  document.getElementById('emerg-panel').style.display='none';
  document.getElementById('sv-island-alert').style.display='none';
  document.getElementById('sv-arrows').style.display='none';
  var nd = document.getElementById('nav-sim-dot');
  if(nd){nd.style.background='var(--amber)';nd.style.boxShadow='0 0 5px var(--amber)';}
  addEqPulses();
  logEvent('T+0:00 — M' + P.magnitude + ' earthquake detected at epicentre.', 'var(--red2)');
  var DURATION = 9000;
  function animate() {
    var elapsed = Date.now() - simStart;
    var t = Math.min(elapsed / DURATION, 1);
    var seisT = Math.min(elapsed / (DURATION * 0.22), 1);
    var tsuT = Math.max(0, Math.min((elapsed - 700) / (DURATION * 0.86), 1));
    seisPos = seisT * 78; tsuPos = tsuT * 58;
    updateDot('sv-swave', seisPos); updateDot('ug-swave', seisPos, true);
    updateDot('cs-swave', seisPos, false, true);
    updateTsuPos(tsuPos);
    updateSeisRings(seisPos);
    var str = document.getElementById('sat-tsunami-ring');
    if (str) str.setAttribute('r', String(tsuPos * 3.8));
    var pfPct = t * 100;
    ['sv-pfill','ug-pfill','cs-pfill'].forEach(function(id){ var e=document.getElementById(id);
if(e)e.style.width=pfPct+'%'; });
    var tlf = document.getElementById('tl-tsunami-fill');
    if (tlf) tlf.style.width = (tsuT * 100) + '%';
    if (t > 0.2 && !sDetected) { sDetected=true; onSeisDetected(); }
    if (t > 0.48 && !bDetected) { bDetected=true; onBuoyDetected(); }
    if (t > 0.54 && !wIssued) { wIssued=true; onWarning(); }
    updateLiveData();
    if (t < 1) simRAF = requestAnimationFrame(animate); else onSimComplete();
  }
}

```



```

    }
    simRAF = requestAnimationFrame(animate);
}

function updateDot(id, pos, ugMode, csMode) {
    var el = document.getElementById(id);
    if (!el) return;
    el.style.left = (15 + pos * 0.58) + '%';
    el.style.display = pos > 0 ? 'block' : 'none';
    if (ugMode) { el.style.bottom = '88px'; } else if (csMode) { el.style.bottom = '82px'; }
}

function updateTsuPos(pos) {
    var pct = 15 + pos * 0.52;
    ['sv-tsunami', 'cs-tsunami'].forEach(function(id){ var el=document.getElementById(id);
    if(el){el.style.left=pct+'%';el.style.display=pos>0?'block':'none';} });
    if (pos > 0) { var a=document.getElementById('sv-arrows'); if(a)a.style.display='block'; }
}

function updateSeisRings(pos) {
    var r = pos * 1.4;
    ['ugr1', 'ugr2', 'ugr3'].forEach(function(id,i){ var el=document.getElementById(id);
    if(el)el.setAttribute('r',String(Math.max(0,r-i*11))); });
}

function addEqPulses() {
    ['surf-pulses', 'ug-pulses', 'cross-pulses'].forEach(function(id){ var
    el=document.getElementById(id); if(!el)return; el.innerHTML='<svg width="48" height="48"
    viewBox="0 0 48 48" style="position:absolute;top:-24px;left:-24px"><circle cx="24" cy="24"
    r="0" fill="none" stroke="var(--red2)" stroke-width="1.8" style="animation:pulseRing 1.8s
    ease-out infinite"/><circle cx="24" cy="24" r="0" fill="none" stroke="var(--red2)"
    stroke-width="1.3" style="animation:pulseRing 1.8s ease-out
    infinite;animation-delay:.55s"/></svg>'; });
}

function activatePanel(id, color, lblId, stId) {
    var panel=document.getElementById(id);
    if(panel){panel.style.borderColor=color;panel.style.boxShadow='0 0 12px
'+color.replace('var(--','rgba(').replace(')','',0.15)');}
    var lbl=document.getElementById(lblId); if(lbl)lbl.style.color=color;
    var st=document.getElementById(stId); if(st){st.textContent='●
ACTIVE';st.style.color=color;st.style.animation='blinkAlert 1s step-end infinite';}
}

function activateSensor(cls, color) {
    document.querySelectorAll('.'+cls).forEach(function(el){
    el.style.background=color.replace('var(--','rgba(').replace(')','',0.15)');el.style.borderColor=colo

```

```

r;el.style.color=color;el.style.boxShadow='0 0 9px
'+color.replace('var(--','rgba(').replace(')','',0.38)); });
}

```

```

function onSeisDetected() {
  activateSensor('sv-sensor','var(--seafoam)');
  activateSensor('ug-sensor','var(--amber)');
  activateSensor('cs-sensor','var(--amber)');

```

```

document.querySelectorAll('.s-status[data-id="s1"],.s-status[data-id="s2"]').forEach(function(
el){ el.textContent='● ACTIVE'; el.style.color='var(--amber)'; });
  activatePanel('db-s','var(--amber)','db-s-lbl','db-s-st');
  activatePanel('db-h','var(--seafoam)','db-h-lbl','db-h-st');
  logEvent('T+' + seisTime().toFixed(1) + 'min — Seismic sensors TRIGGERED! P-wave
confirmed.', 'var(--amber)');
}

```

```

function onBuoyDetected() {
  document.querySelectorAll('.buoy-top,.cs-buoy-top').forEach(function(el){
try{el.setAttribute('stroke','var(--blue)');el.setAttribute('fill','rgba(59,130,246,.3)');}catch(e){} });
  document.querySelectorAll('.buoy-lbl').forEach(function(el){ el.style.color='var(--blue)'; });
  [sat-b1,sat-b2].forEach(function(id){ var e=document.getElementById(id);
if(e){e.setAttribute('fill','rgba(59,130,246,.8)');e.setAttribute('stroke','white');} });

```

```

document.querySelectorAll('.s-status[data-id="b1"],.s-status[data-id="b2"]').forEach(function(
el){ el.textContent='● ACTIVE'; el.style.color='var(--blue)'; });
  activatePanel('db-b','var(--blue)','db-b-lbl','db-b-st');
  logEvent('T+' + (seisTime() * 3).toFixed(0) + 'min — DART buoys confirm tsunami wave!',
'var(--blue)');
}

```

```

function onWarning() {
  document.getElementById('sv-island-alert').style.display='block';
  document.getElementById('sv-island').style.animation='shake 0.4s ease-in-out infinite';
  document.querySelectorAll('.sv-building').forEach(function(b){
b.setAttribute('fill','var(--red2)'); });
  var sat=document.getElementById('sat-warn'); if(sat)sat.style.display='block';
  var ctrl=document.getElementById('sv-ctrl');
if(ctrl){ctrl.style.borderColor='var(--red2)';ctrl.style.background='rgba(220,38,38,.08)';ctrl.style
.boxShadow='0 0 14px rgba(220,38,38,.25)';}
  var cl=document.getElementById('sv-ctrl-lbl'); if(cl)cl.style.color='var(--red2)';
  document.querySelectorAll('.s-status[data-id="warn"]').forEach(function(el){
el.textContent='● ACTIVE'; el.style.color='var(--red2)'; el.style.animation='blinkAlert .8s
step-end infinite'; });
  activatePanel('db-a','var(--red2)','db-a-lbl','db-a-st');
  document.getElementById('emerg-panel').style.display='block';

```

```

    var nd=document.getElementById('nav-sim-dot');
    if(nd){nd.style.background='var(--red2)';nd.style.boxShadow='0 0 6px
    var(--red2)';nd.style.animation='blinkAlert .8s step-end infinite';}
    logEvent('T+' + (seisTime() * 3.2).toFixed(0) + 'min — ⚠ TSUNAMI WARNING ISSUED
    FOR WAVARA!', 'var(--red2)');
}

```

```

function onSimComplete() {
    simState='complete';
    document.getElementById('sim-btn').disabled=false;
    document.getElementById('sim-btn').textContent='⚡ Start Earthquake';
    document.getElementById('sv-complete').style.display='block';
    logEvent('SIMULATION COMPLETE — Hybrid advantage: ' + hybridAdv().toFixed(1) + '
    hrs.', 'var(--green)');
}

```

```

function logEvent(txt, color) {
    var log=document.getElementById('event-log');
    var d=document.createElement('div');
    d.style.color=color||'var(--text)'; d.style.animation='dashIn .3s ease'; d.textContent=txt;
    log.appendChild(d); log.scrollTop=log.scrollHeight;
    while(log.children.length>10) log.removeChild(log.firstChild);
}

```

```

function resetSim() {
    if(simRAF){cancelAnimationFrame(simRAF);simRAF=null;}
    simState='idle';sDetected=bDetected=wIssued=false;seisPos=tsuPos=0;
    document.getElementById('sim-btn').disabled=false;
    document.getElementById('sim-btn').textContent='⚡ Start Earthquake';
    ['sv-pbar','ug-pbar','cs-pbar'].forEach(function(id){var
    e=document.getElementById(id);if(e)e.style.display='none';});
    ['sv-pfill','ug-pfill','cs-pfill'].forEach(function(id){var
    e=document.getElementById(id);if(e)e.style.width='0';});
    document.getElementById('sv-complete').style.display='none';
    document.getElementById('emerg-panel').style.display='none';
    document.getElementById('sv-island-alert').style.display='none';
    document.getElementById('sv-island').style.animation='';
    document.getElementById('sv-arrows').style.display='none';
    ['sv-swave','ug-swave','cs-swave'].forEach(function(id){var
    e=document.getElementById(id);if(e)e.style.display='none';});
    ['sv-tsunami','cs-tsunami'].forEach(function(id){var
    e=document.getElementById(id);if(e)e.style.display='none';});
    var str=document.getElementById('sat-tsunami-ring'); if(str)str.setAttribute('r','0');
    var sw=document.getElementById('sat-warn'); if(sw)sw.style.display='none';
    ['sv-sensor','ug-sensor','cs-sensor'].forEach(function(cls){
    document.querySelectorAll('.'+cls).forEach(function(el){el.style.background='var(--panel2)';el
    .style.borderColor='var(--muted2)';el.style.color='var(--muted2)';el.style.boxShadow='none';})
    ; });
}

```

```

    document.querySelectorAll('.buoy-top,.cs-buoy-top').forEach(function(el){
try{el.setAttribute('stroke','var(--muted2)');el.setAttribute('fill','rgba(27,42,58,.8)');}catch(e){} });
    document.querySelectorAll('.buoy-lbl').forEach(function(el){el.style.color='var(--muted2)');});
    ['sat-b1','sat-b2'].forEach(function(id){var
e=document.getElementById(id);if(e){e.setAttribute('fill','rgba(91,192,190,.65)');e.setAttribute(
'stroke','var(--seafoam)');});
    document.querySelectorAll('.s-status').forEach(function(el){el.textContent='○
Standby';el.style.color='var(--border2)';el.style.animation=";});
    ['db-s','db-b','db-h','db-a'].forEach(function(id){var
e=document.getElementById(id);if(e){e.style.borderColor='var(--border)';e.style.background=
'var(--bg)';e.style.boxShadow='none';});
    ['db-s-lbl','db-b-lbl','db-h-lbl','db-a-lbl'].forEach(function(id){var
e=document.getElementById(id);if(e)e.style.color='var(--muted2)');});
    ['db-s-st','db-b-st','db-h-st','db-a-st'].forEach(function(id){var
e=document.getElementById(id);if(e){e.textContent='○
Standby';e.style.color='var(--border2)';e.style.animation=";});
    var ctrl=document.getElementById('sv-ctrl');
if(ctrl){ctrl.style.borderColor='var(--border)';ctrl.style.background='var(--panel)';ctrl.style.boxS
hadow='none';}
    var cl=document.getElementById('sv-ctrl-lbl'); if(cl)cl.style.color='var(--muted2)';

document.querySelectorAll('.sv-building').forEach(function(b){b.setAttribute('fill','#1e3040');});
    ['surf-pulses','ug-pulses','cross-pulses'].forEach(function(id){var
e=document.getElementById(id);if(e)e.innerHTML=";});
    var nd=document.getElementById('nav-sim-dot');
if(nd){nd.style.background='var(--border2)';nd.style.boxShadow='none';nd.style.animation=";}
    document.getElementById('event-log').innerHTML='<div>System standby...</div>';
    var tlf=document.getElementById('tl-tsunami-fill'); if(tlf)tlf.style.width='0';
}

/* ===== WAVE SCIENCE ===== */
function updateWaveScience() {
    var f=parseFloat(document.getElementById('ws-f').value);
    var a=parseFloat(document.getElementById('ws-a').value);
    setText('ws-fv',f.toFixed(1)+' Hz');
    setText('ws-av',a);
    setText('ws-formula','v = fλ · f = '+f+' Hz · T = '(1/f).toFixed(2)+' s');
}

function startWaveAnim() {
    if(waveRAF)return;
    var t=0;
    function frame() {
        t+=0.038;
        var f=parseFloat(document.getElementById('ws-f').value)||1;
        var a=parseFloat(document.getElementById('ws-a').value)||30;
        var W=560,H=100;
        var pts=[];

```

```

for(var x=0;x<=W;x+=2){ var y=H/2+a*Math.sin((x/W)*f*Math.PI*4-t*f); pts.push(x+', '+y); }
var wp=document.getElementById('ws-path'); if(wp)wp.setAttribute('d','M '+pts.join(' L '));
var IPx=W/(f*2);
var wLine=document.getElementById('ws-wl-line'); if(wLine)wLine.setAttribute('x2',IPx);
var wTxt=document.getElementById('ws-wl-txt'); if(wTxt)wTxt.setAttribute('x',IPx/2);
var crest=document.getElementById('ws-crest');
if(crest){crest.setAttribute('cx',W/(f*4));crest.setAttribute('cy',H/2-a);}
// Transverse
var tvPts=[];
for(var i=0;i<16;i++){var xi=20+i*32;var yi=48+26*Math.sin(i*0.7-t*2);tvPts.push(xi+', '+yi);}
var tvp=document.getElementById('tv-path'); if(tvp)tvp.setAttribute('d','M '+tvPts.join(' L '));
// Longitudinal
var lg=document.getElementById('lv-dots');
if(lg){var s="";for(var i2=0;i2<20;i2++){var bx=14+i2*24;var
dx=10*Math.sin(i2*0.8-t*2);s+=''<circle cx=""+(bx+dx)+" cy="48" r="4" fill="--amber"
fill-opacity=".75"/>';}lg.innerHTML=s;}
waveRAF=requestAnimationFrame(frame);
}
frame();
}
function stopWaveAnim(){if(waveRAF){cancelAnimationFrame(waveRAF);waveRAF=null;}}

```

/* ===== EQUATION ===== */

```

function updateEq() {
var f=parseFloat(document.getElementById('eq-f').value);
var wl=parseFloat(document.getElementById('eq-w').value);
var spd=Math.round(f*wl);
setText('eq-fv',f.toFixed(1)+' Hz'); setText('eq-wv',wl+' m');
setText('eq-f-disp',f.toFixed(1)+' Hz'); setText('eq-w-disp',wl+' m');
setText('eq-spd-disp',spd+' m/s');
setText('eq-wl-ann','λ = '+wl+'m');
setText('eq-formula','v = f × λ = '+f+' × '+wl+' = '+spd+' m/s · T = 1/f = '+ (1/f).toFixed(3)+'
s');
}
function startEqAnim() {
if(eqRAF)return;
var t=0;
function frame() {
t+=0.05;
var f=parseFloat(document.getElementById('eq-f').value)||2;
var wl=parseFloat(document.getElementById('eq-w').value)||80;
var W=560,H=115;
var k=(2*Math.PI)/wl;
var omega=2*Math.PI*f;
var pts=[];
for(var x=0;x<=W;x+=2){var y=H/2+34*Math.sin(k*x-omega*t*0.02);pts.push(x+', '+y);}
var ep=document.getElementById('eq-path'); if(ep)ep.setAttribute('d','M '+pts.join(' L '));
var wll=document.getElementById('eq-wl-line'); if(wll)wll.setAttribute('x2',20+wl);
}
}

```

```

    var wla=document.getElementById('eq-wl-ann'); if(wla)wla.setAttribute('x',20+wl/2);
    eqRAF=requestAnimationFrame(frame);
  }
  frame();
}
function stopEqAnim(){if(eqRAF){cancelAnimationFrame(eqRAF);eqRAF=null;}}

/* ===== CONCEPT CARDS ===== */
function showC(el,title,desc){el.style.borderColor='var(--seafoam2)';el.style.boxShadow='0 0
16px rgba(91,192,190,.1)';var
b=el.querySelector('.cc-body');if(b){b.style.fontSize='9.5px';b.style.color='var(--text)';b.style.li
neHeight='1.65';b.style.animation='fadeIn .2s ease';b.textContent=desc;}}
function hideC(el){el.style.borderColor='var(--border)';el.style.boxShadow='none';var
b=el.querySelector('.cc-body');if(b){b.style.fontSize='8.5px';b.style.color='var(--muted2)';b.tex
tContent='Hover →';}}

/* ===== COMPARISON ===== */
var CMP={
  seismic:{label:'System A — Seismic
Only',color:'var(--seafoam2)',stats:{cost:3,speed:9,reliability:7,falseAlarm:5,evacuation:9},des
c:'Seismic sensors detect earthquakes instantly but cannot confirm if a tsunami was
generated. High speed, elevated false alarm risk.'},
  buoy:{label:'System B — Buoy
Only',color:'var(--blue)',stats:{cost:5,speed:5,reliability:8,falseAlarm:3,evacuation:5},desc:'DA
RT buoys confirm tsunami presence with high accuracy — but the wave must reach the
buoy first, destroying the warning time advantage.'},
  hybrid:{label:'★ Hybrid
WaveGuard',color:'var(--green)',stats:{cost:7,speed:8,reliability:10,falseAlarm:2,evacuation:9
},desc:'Seismic speed + buoy confirmation accuracy. 6.5 hour warning advantage. Minimum
false alarms. Superior across every metric.'}
};
var CMP_METRICS=[{key:'speed',label:'Detection
Speed'},{key:'reliability',label:'Reliability'},{key:'evacuation',label:'Evacuation
Advantage'},{key:'falseAlarm',label:'False Alarm
Risk',invert:true},{key:'cost',label:'Implementation Cost',invert:true}];
var selectedSys='seismic';

function selectSys(key) {
  selectedSys=key;
  var s=CMP[key];
  var t=document.getElementById('cmp-title'); if(t){t.textContent=s.label;t.style.color=s.color;}
  setText('cmp-desc',s.desc);
  var m=document.getElementById('cmp-metrics');
  if(m){m.innerHTML="";CMP_METRICS.forEach(function(mt){var val=s.stats[mt.key];var
disp=mt.invert?10-val:val;m.innerHTML+=`<div style="margin-bottom:9px"><div
style="display:flex;justify-content:space-between;margin-bottom:2px"><span class="mono"
style="font-size:9.5px;color:var(--muted2)">'+mt.label+'</span><span class="mono"
style="font-size:10.5px;color:'+s.color+'">'+disp+'</span></div><div class="bar"><div

```

```

class="bar-fill"
style="width:'+(disp*10)+'%;background:linear-gradient(90deg,var(--accent),'+s.color+');transition:width .5s ease"></div></div></div>');});
    ['seismic','buoy','hybrid'].forEach(function(k){var
    btn=document.getElementById('cmp-'+k[0]+'-btn') ||
    document.getElementById('cmp-'+(k==='seismic'?s:k==='buoy'?b:'h')+'-btn');});
    document.getElementById('cmp-s-btn').className='tab-btn'+(key==='seismic'? active:'');
    document.getElementById('cmp-b-btn').className='tab-btn'+(key==='buoy'? active:'');
    document.getElementById('cmp-h-btn').className='tab-btn'+(key==='hybrid'? active:'');
    updateRadar();
}

```

```

function pentagon(cx,cy,r){return Array.from({length:5},function(_,i){var
a=(i/5)*Math.PI*2-Math.PI/2;return (cx+r*Math.cos(a))+','+(cy+r*Math.sin(a));}).join(' ');}
function radarPts(cx,cy,vals,maxR){var n=vals.length;return vals.map(function(v,i){var
a=(i/n)*Math.PI*2-Math.PI/2;var r=(v/10)*maxR;return
(cx+r*Math.cos(a))+','+(cy+r*Math.sin(a));}).join(' ');}

```

```

function initRadar(){[2,4,6,8,10].forEach(function(lvl){var
el=document.getElementById('rp'+lvl);if(el)el.setAttribute('points',pentagon(150,118,lvl/10*86));});}
function updateRadar(){
    Object.entries(CMP).forEach(function(e){var key=e[0],sys=e[1];var
    el=document.getElementById('radar-'+key);if(!el)return;var
    vals=[sys.stats.speed,sys.stats.reliability,sys.stats.evacuation,10-sys.stats.falseAlarm,10-sys
    .stats.cost];el.setAttribute('points',radarPts(150,118,vals,86));var
    isSel=key===selectedSys;el.setAttribute('stroke-width',isSel?'2.3':'1');el.setAttribute('stroke-o
    pacity',isSel?'1':'.45');});
}

```

/* ===== GRAPHS ===== */

```

function updateGraphs(){
    var depths=[500,1000,2000,3000,4000,5000,6000,7000,8000,10000,12000];
    var speeds=depths.map(function(d){return Math.round(Math.sqrt(9.81*d)*3.6);});
    var maxSpd=Math.max.apply(null,speeds);
    var W=260,H=162,pL=35,pB=27,pT=11,pR=8;
    var gW=W-pL-pR,gH=H-pB-pT;
    var dToX=function(d){return pL+((d-500)/(12000-500))*gW;};
    var sToY=function(s){return pT+gH-(s/maxSpd)*gH;};
    var lp=depths.map(function(d,i){return (i===0?'M':'L')+ ' '+dToX(d)+'
'+sToY(speeds[i]);}).join(' ');
    var pathEl=document.getElementById('g-depth-path'); if(pathEl)pathEl.setAttribute('d',lp);
    var curX=dToX(Math.min(P.oceanDepth,12000));
    var curSpd=Math.round(Math.sqrt(9.81*P.oceanDepth)*3.6);
    var curY=sToY(Math.min(curSpd,maxSpd));
    var vl=document.getElementById('g-depth-vline');
    if(vl){vl.setAttribute('x1',curX);vl.setAttribute('x2',curX);}
}

```

```

    var dot=document.getElementById('g-depth-dot');
    if(dot){dot.setAttribute('cx',curX);dot.setAttribute('cy',curY);}
    setText('g-depth-lbl',P.oceanDepth+'m → '+curSpd+' km/h');
    var g=document.getElementById('g-speed-bars');
    if(g){
        var items=[{lbl:'Seismic Wave',val:P.seismicSpeed,color:'var(--amber)},{lbl:'Tsunami
Wave',val:P.tsunamiSpeed,color:'var(--blue)},{lbl:'Sound
(Water)',val:5400,color:'var(--muted2)'}];
        var maxV=Math.max.apply(null,[P.seismicSpeed,P.tsunamiSpeed,5400,1]);
        g.innerHTML=items.map(function(it,i){var bw=Math.max(4,(it.val/maxV)*215);var
y=18+i*44;var vs=it.val>=1000?(it.val/1000).toFixed(1)+'k':it.val;return '<text x="9" y="'+y+"
fill="'+it.color+" font-size="8.5" font-family='\"Courier New\",monospace">'+it.lbl+'</text><rect
x="9" y="'+(y+5)+" width="'+bw+" height="15" rx="2" fill="'+it.color+" fill-opacity=".25"/><rect
x="9" y="'+(y+5)+" width="'+bw+" height="2.5" rx="1" fill="'+it.color+"><text x="'+(13+bw)+"
y="'+(y+17)+" fill="'+it.color+" font-size="7.5" font-family='\"Courier
New\",monospace">'+vs+' km/h</text>'}).join("");
        var ratio=Math.round(P.seismicSpeed/P.tsunamiSpeed);
        setText('g-speed-note','Seismic ~'+ratio+'× faster than tsunami');
    }
}

/* ===== INIT ===== */
function initApp(){
    genStars('hero-stars',42);
    updateParams(); updateLiveData(); initRadar(); selectSys('seismic'); updateGraphs();
    document.getElementById('safety-date').textContent=new Date().toUTCString();
}
</script>
</body>
</html>

```